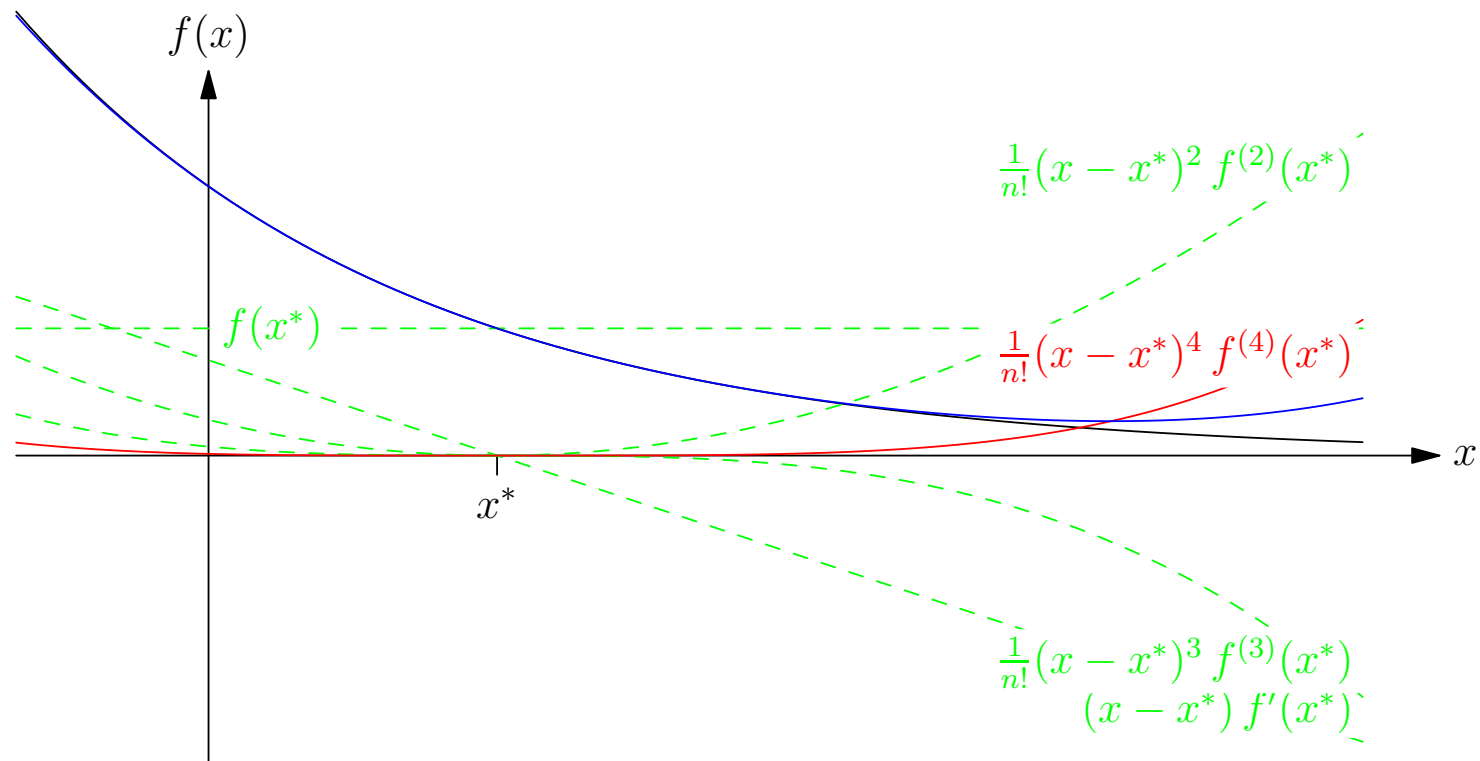


Mathematics of Machine Learning

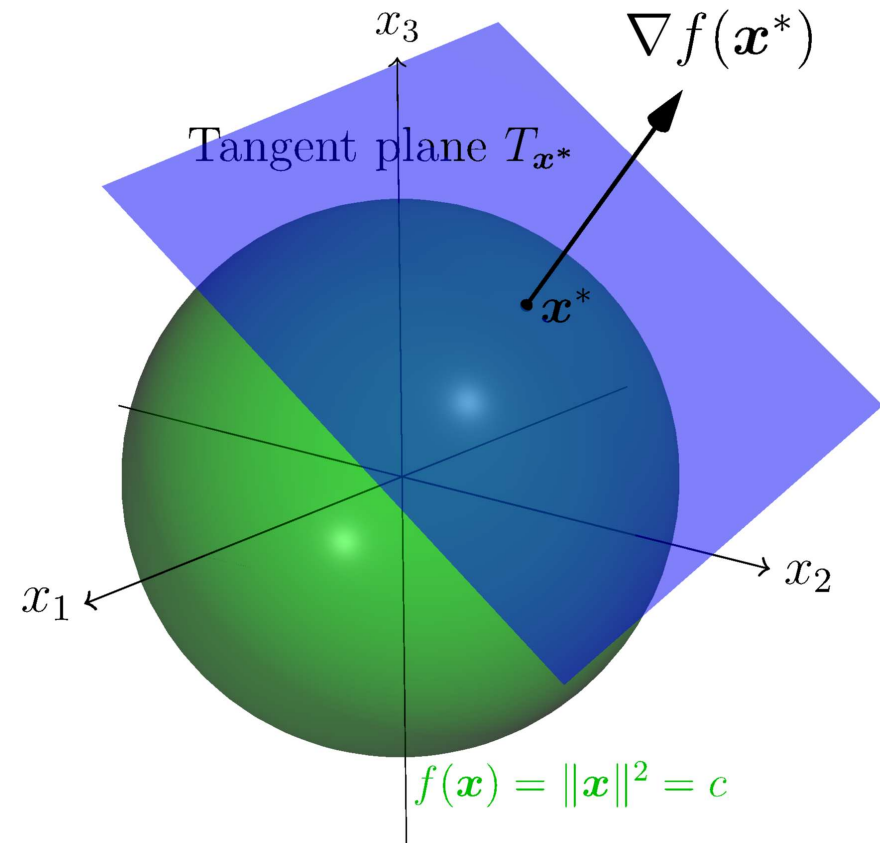
Taylor's Expansion



Convergence, remainder, high-dimensions, Hessian

Outline

1. **1-D Taylor Expansion**
2. High Dimensions
3. Hessians



Approximating Functions

- In much of machine learning we are trying to approximate functions
- There are many different ways to do this
- Depending on your application some are much better than others
- All function approximation makes some assumptions about the regularity of the function they are approximating
- Taylor expansions are polynomial approximations of a smooth function around a point

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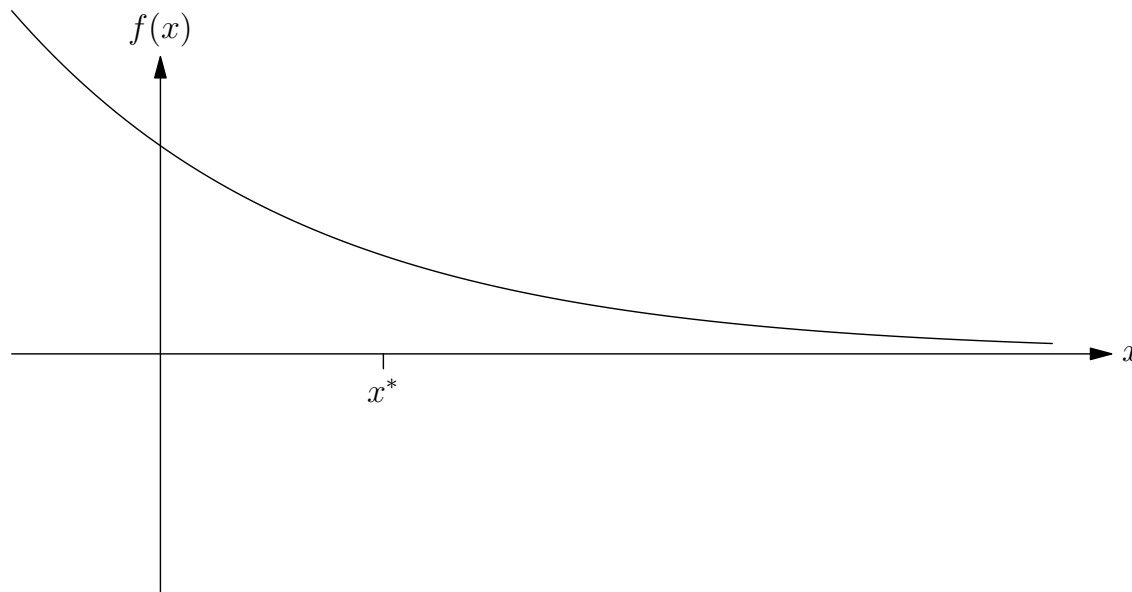
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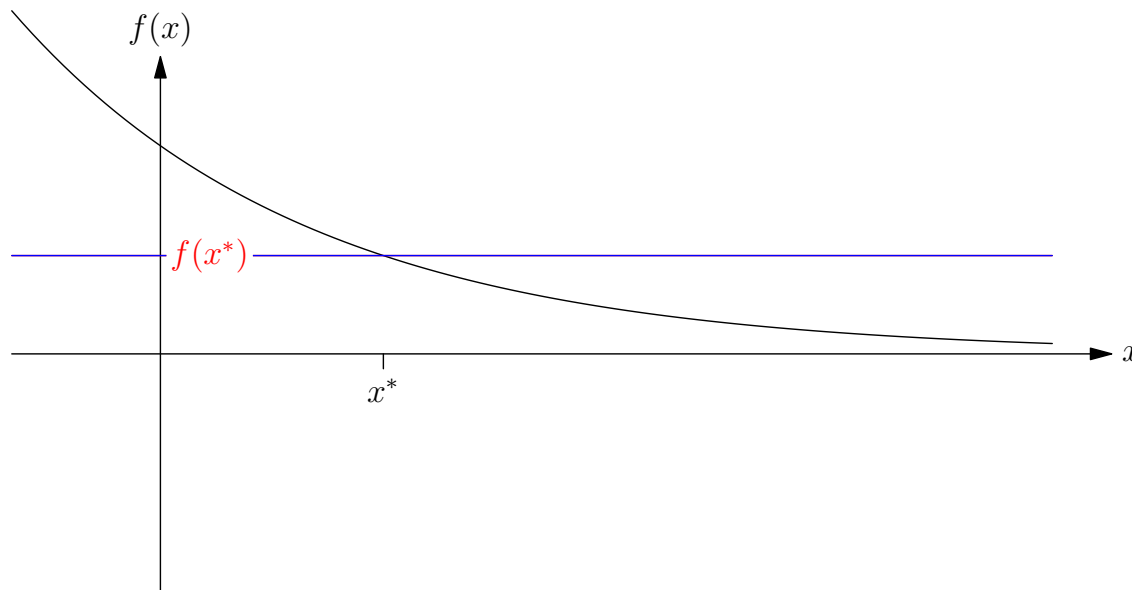


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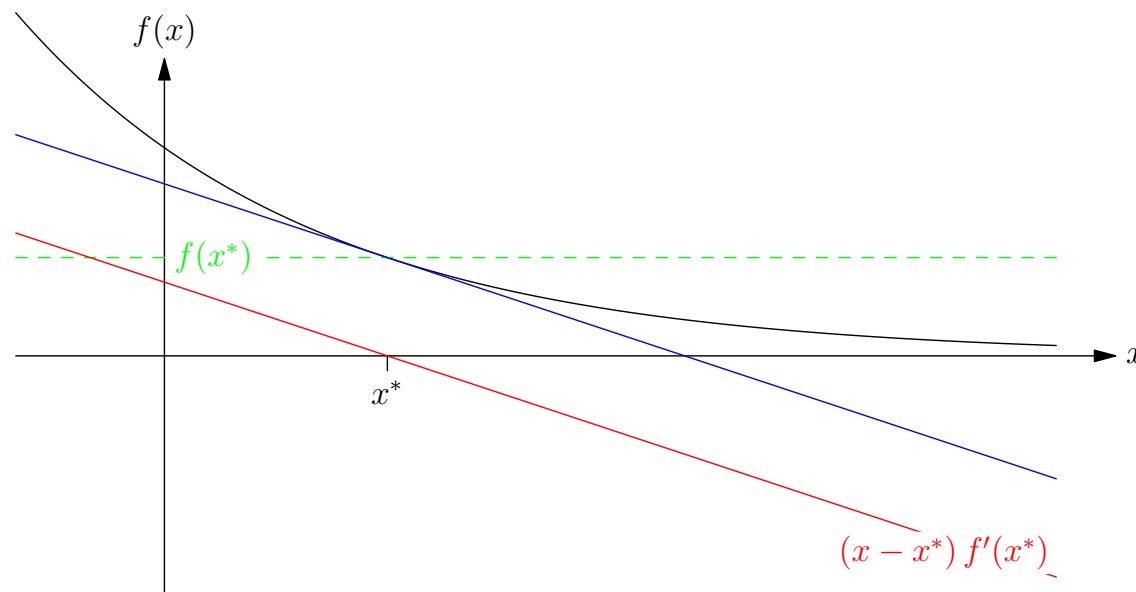


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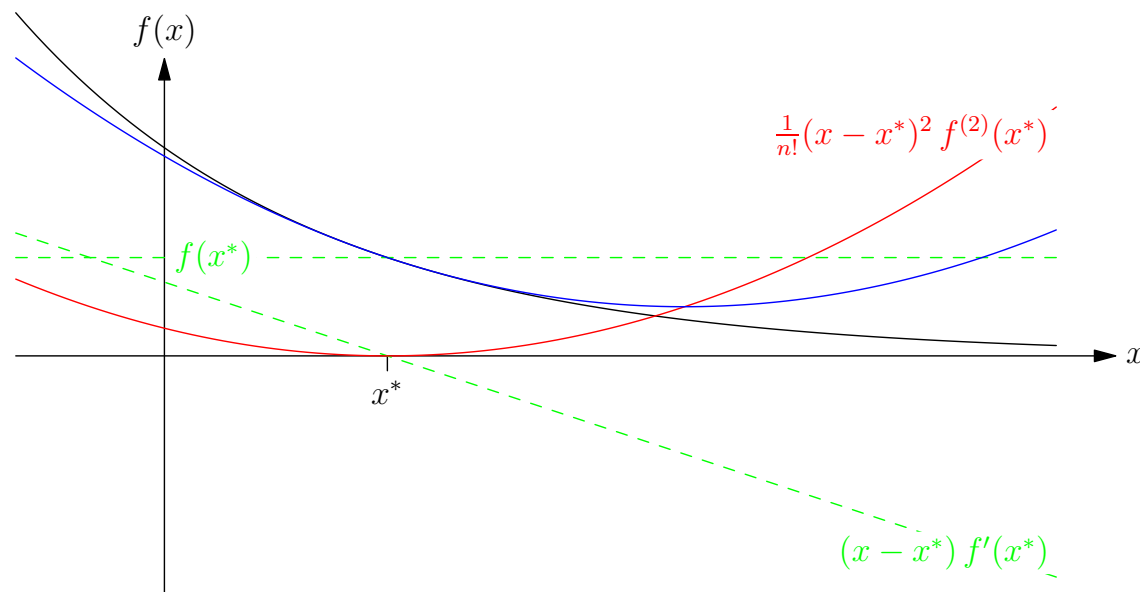


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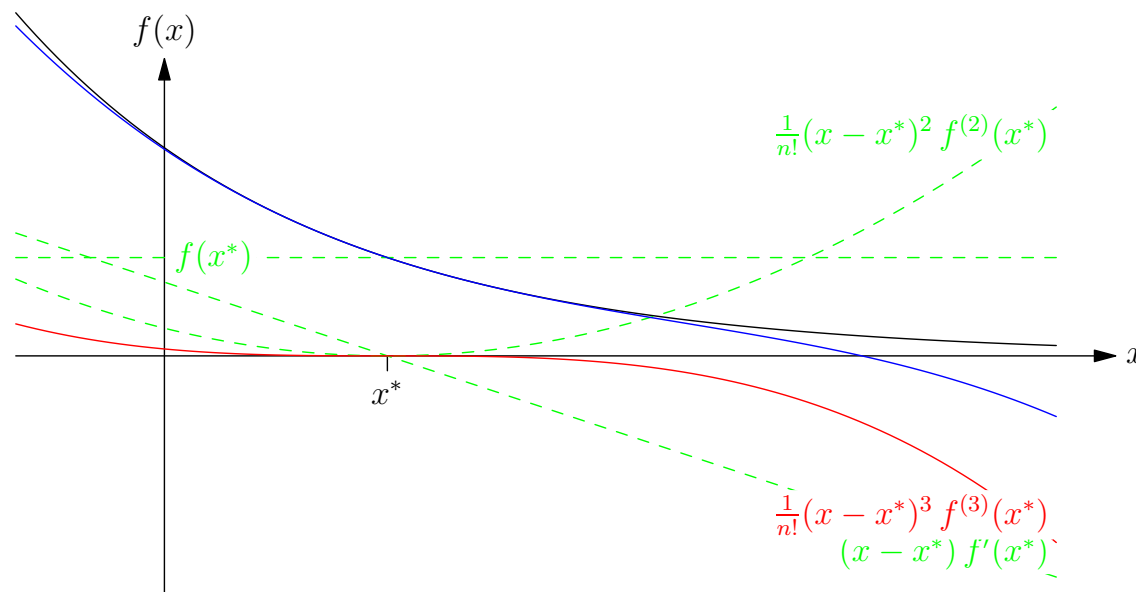


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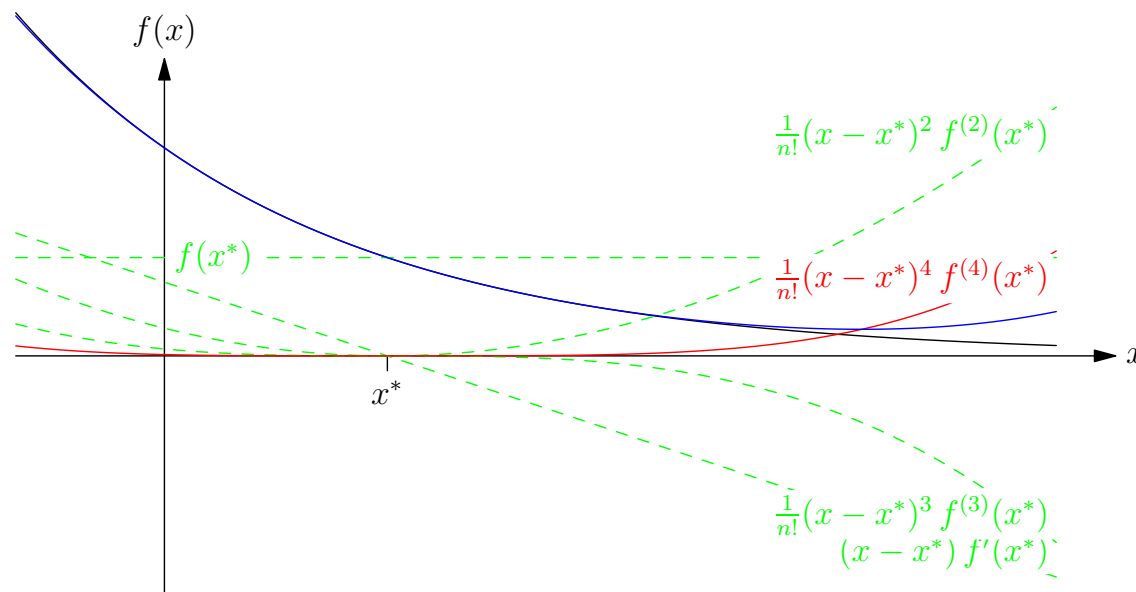


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Convergence

- As we add more and more terms our local approximation becomes more accurate
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- However, polynomials cannot capture functions that diverge for finite numbers, e.g.

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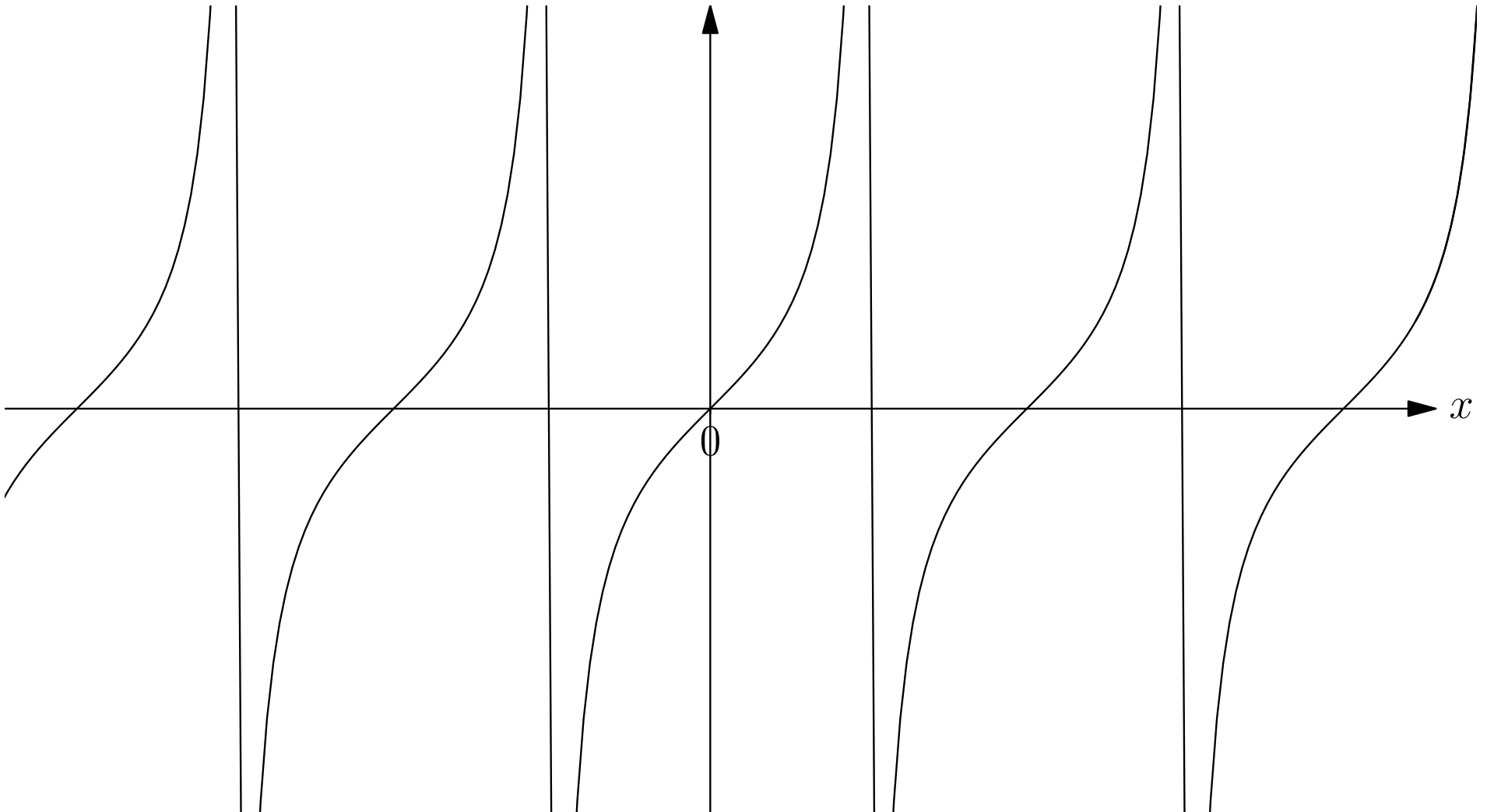
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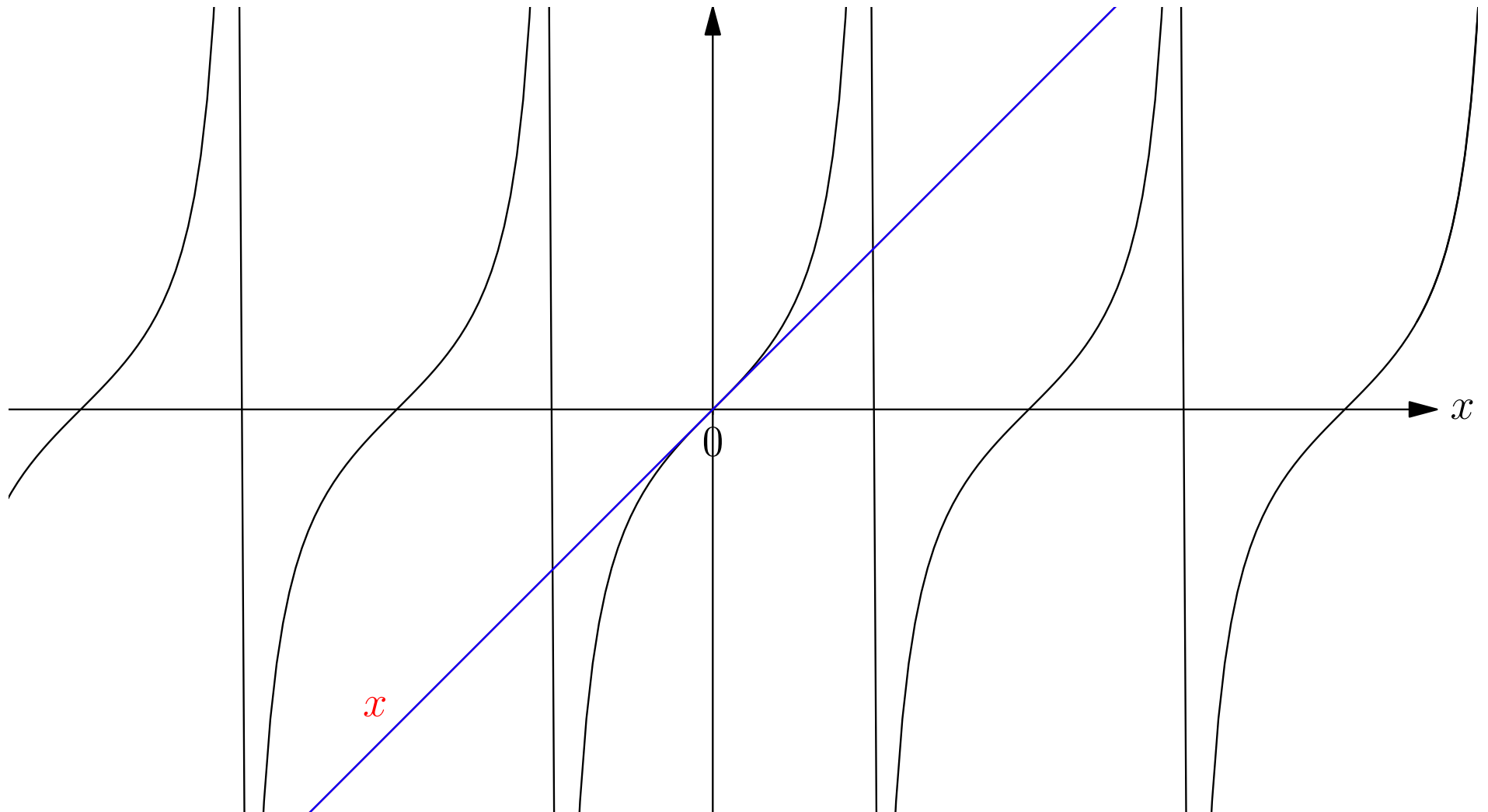
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- This leads to a loss of global convergence

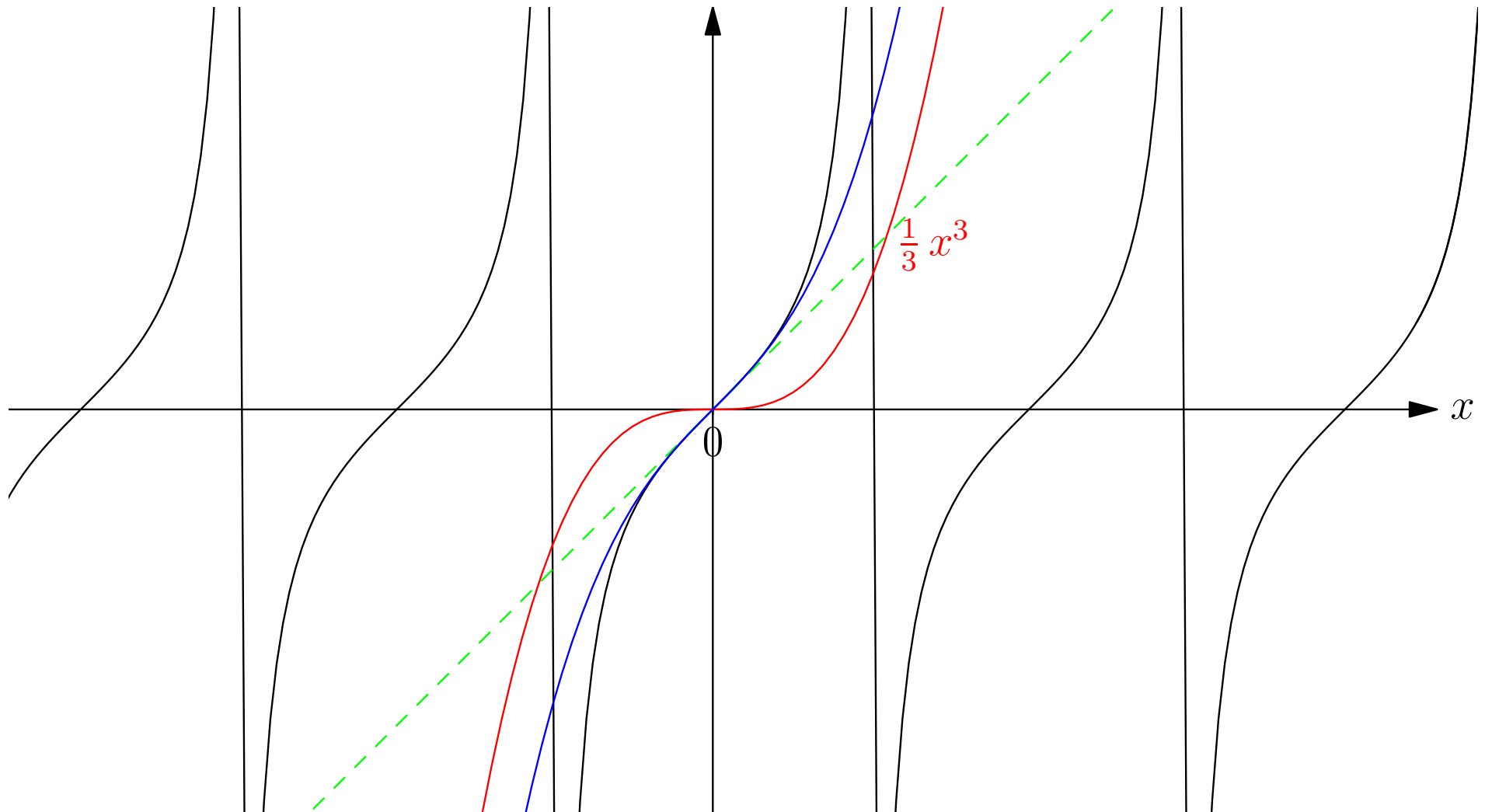
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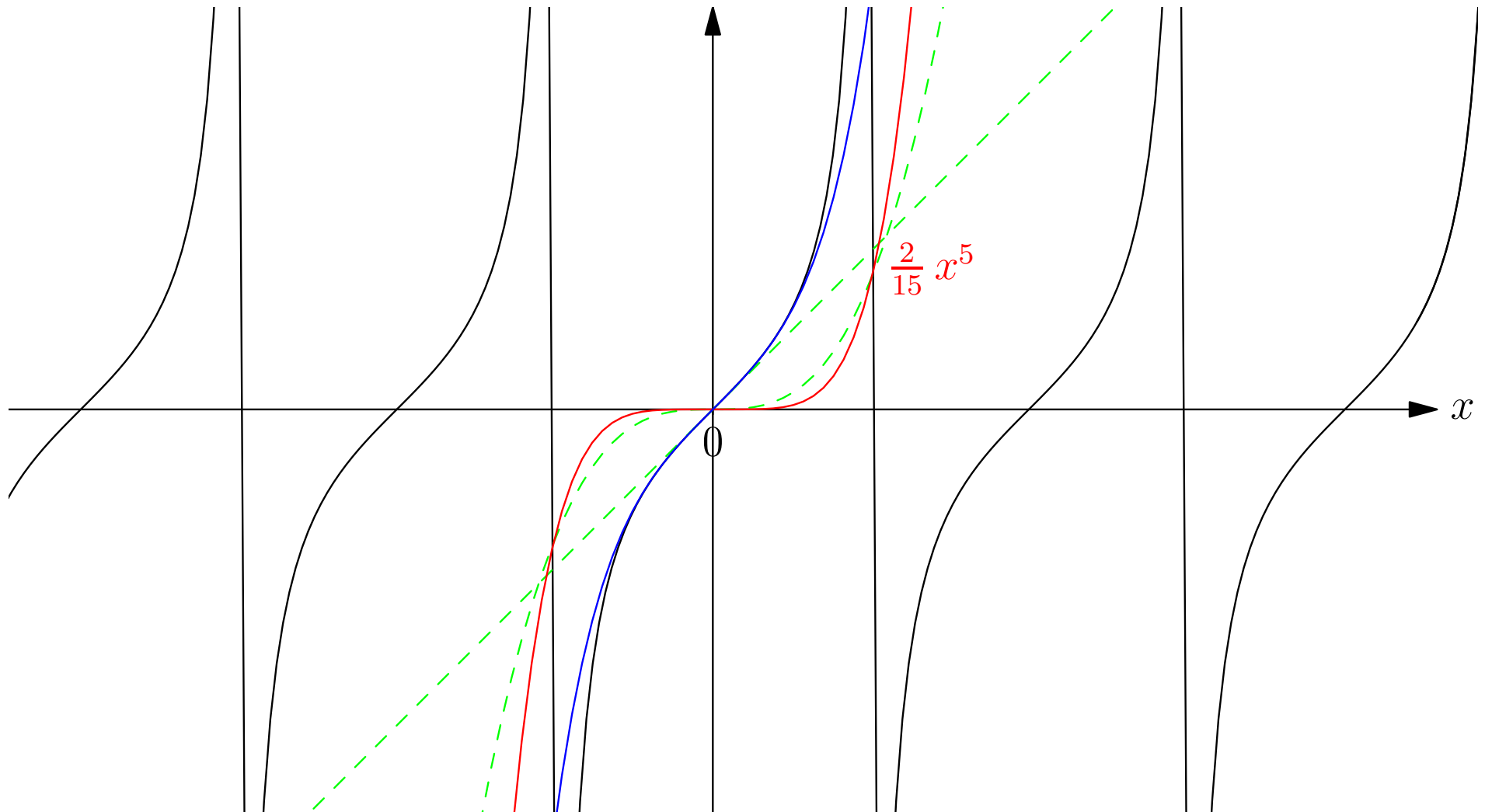
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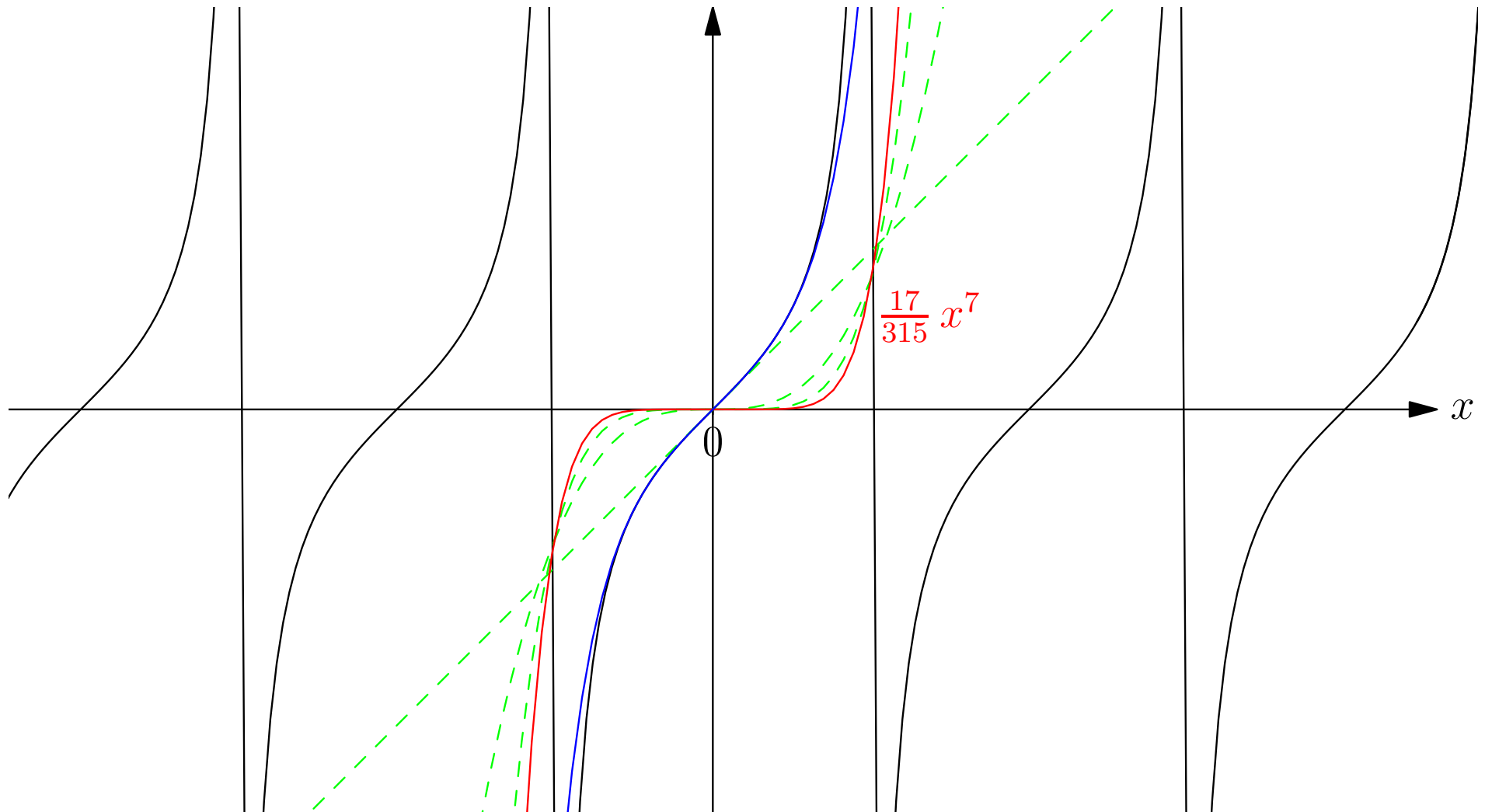
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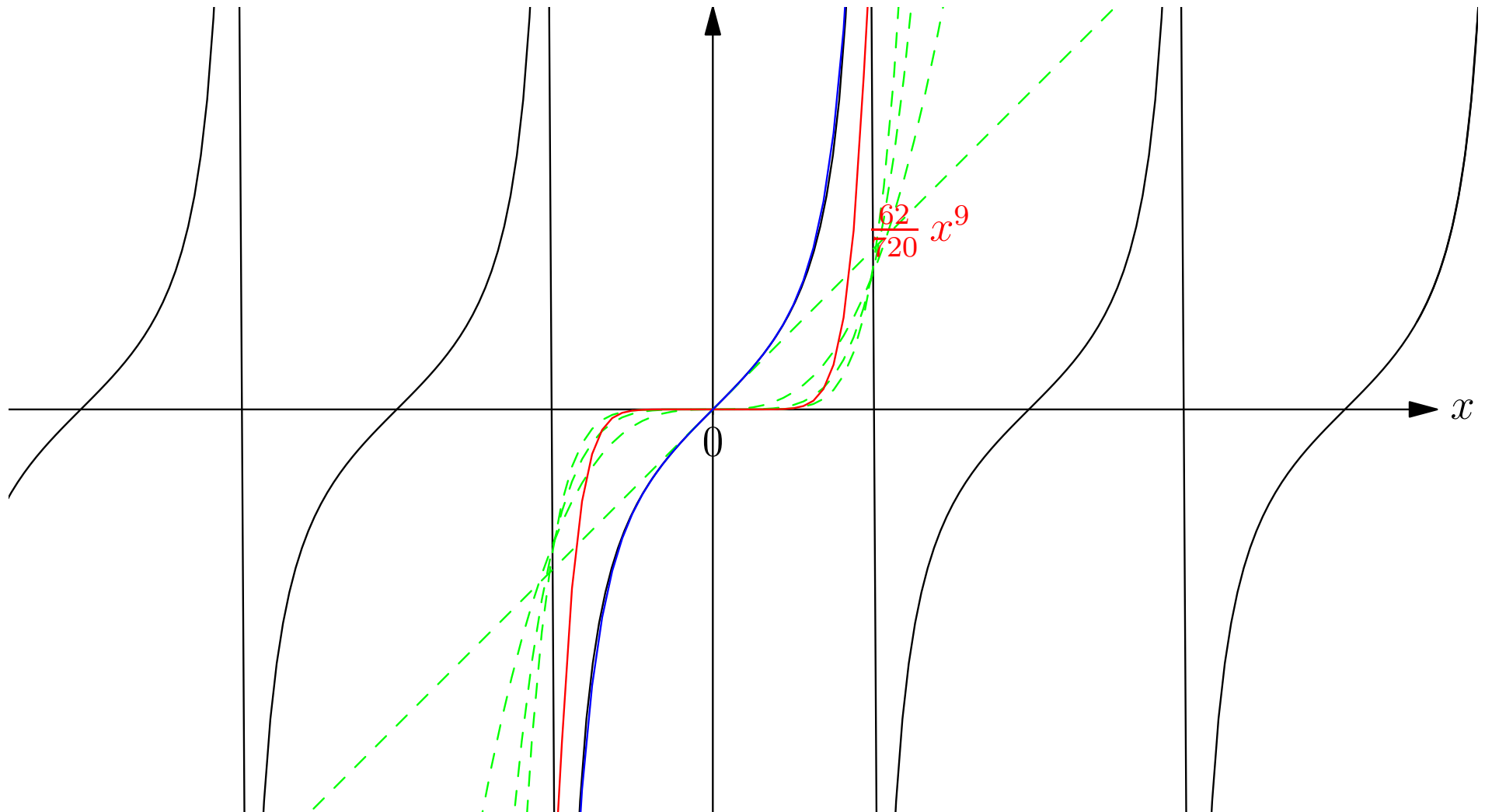
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Complex Functions

- To understand convergence of Taylor series more generally we need to go into the complex plane
- We can generalise many functions by considering them as a function with a complex argument $z = x + iy$
- The value of the complex function will be complex valued
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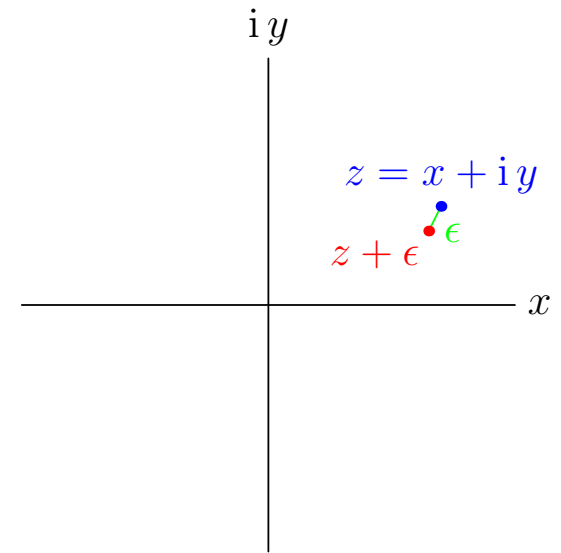
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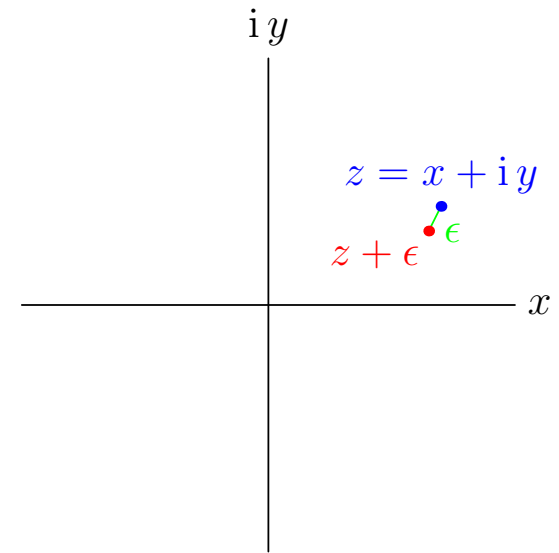
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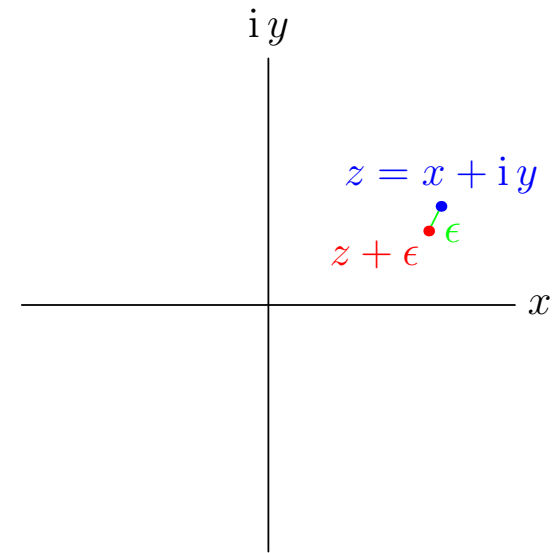
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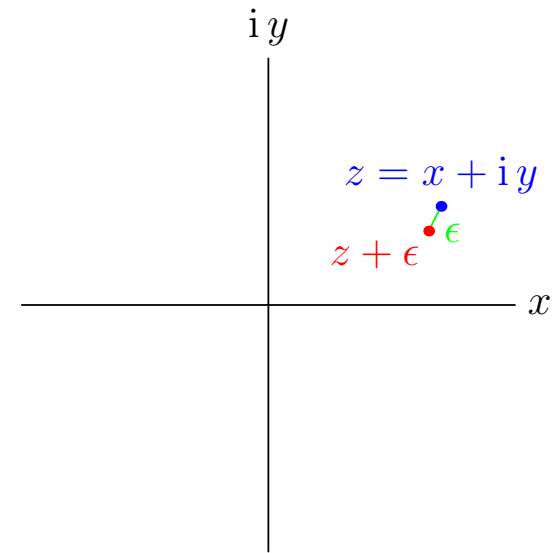
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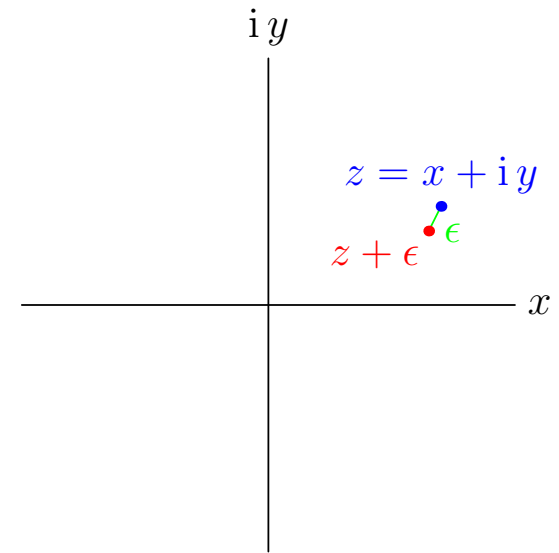
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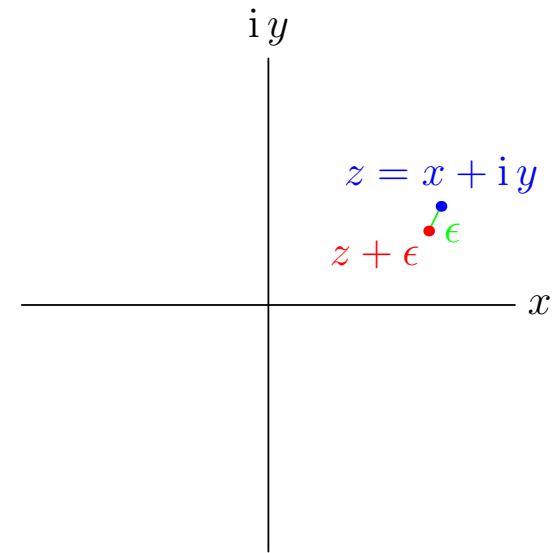
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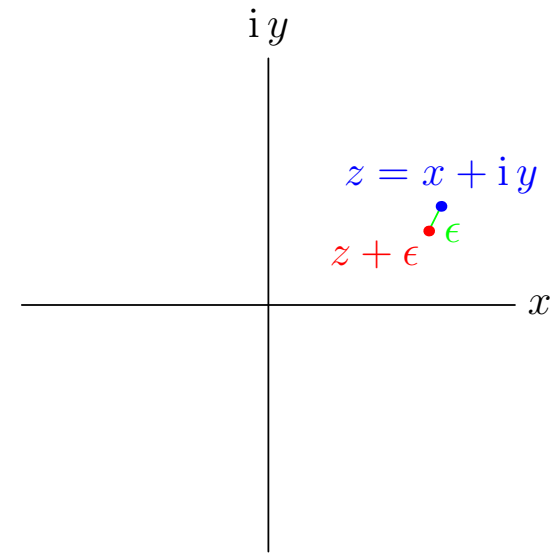
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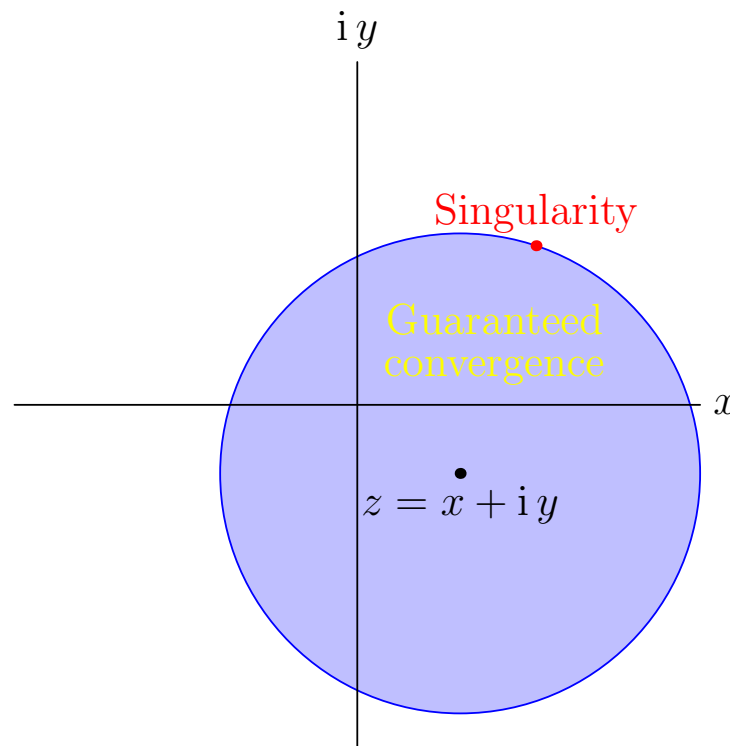


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Guaranteed Convergence of Taylor Series

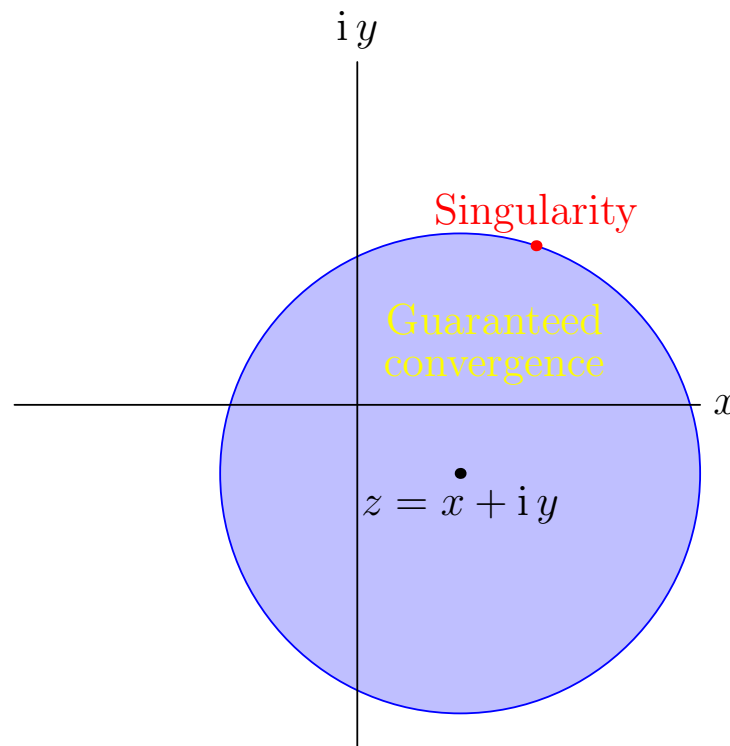
- For analytic functions, Taylor series are guaranteed to converge inside a circle that excludes singular points



- E.g. the function $f(z) = \frac{1}{z^2+1}$ has a singularity at $z = \pm i$

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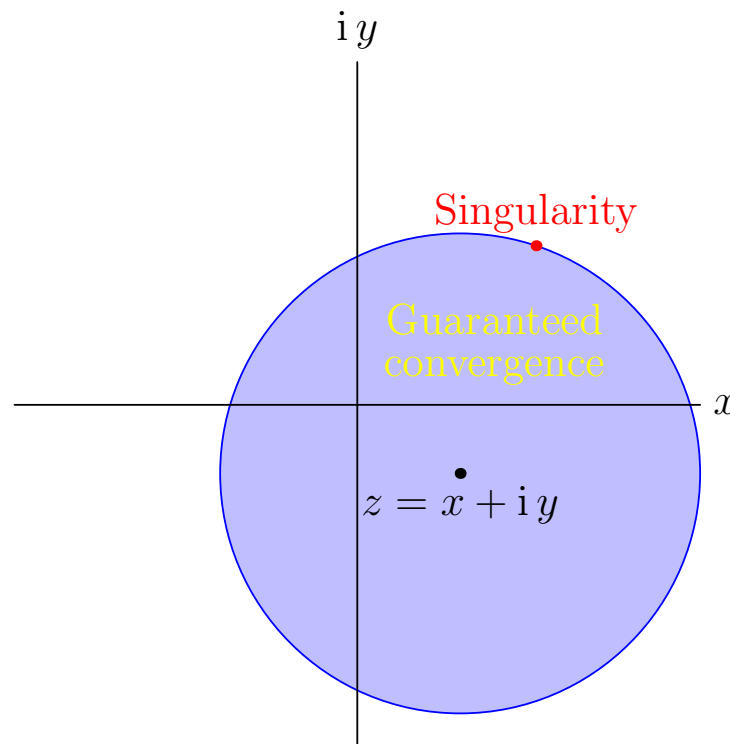
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Guaranteed Convergence of Taylor Series

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- E.g. the function $f(z) = \frac{1}{z^2+1}$ has a singularity at $z = \pm i$. The Taylor series of $f(z)$ around 0 is guaranteed to converge inside a circle of radius 1

The Hyperbolic Tangent

- The hyperbolic tangent, $\tanh$, is related to the normal tangent function through $\tanh(x) = -i \tan(ix)$
- The function (along the real line) doesn't have any singularities
- But we know that $\tan(x)$ has singularities at $\frac{\pi}{2} + k\pi$ where k is any integer, so $\tanh(x)$ has singularities at $x = \pm i\frac{\pi}{2}, \pm i\frac{3\pi}{2}, \dots$ on the imaginary axis
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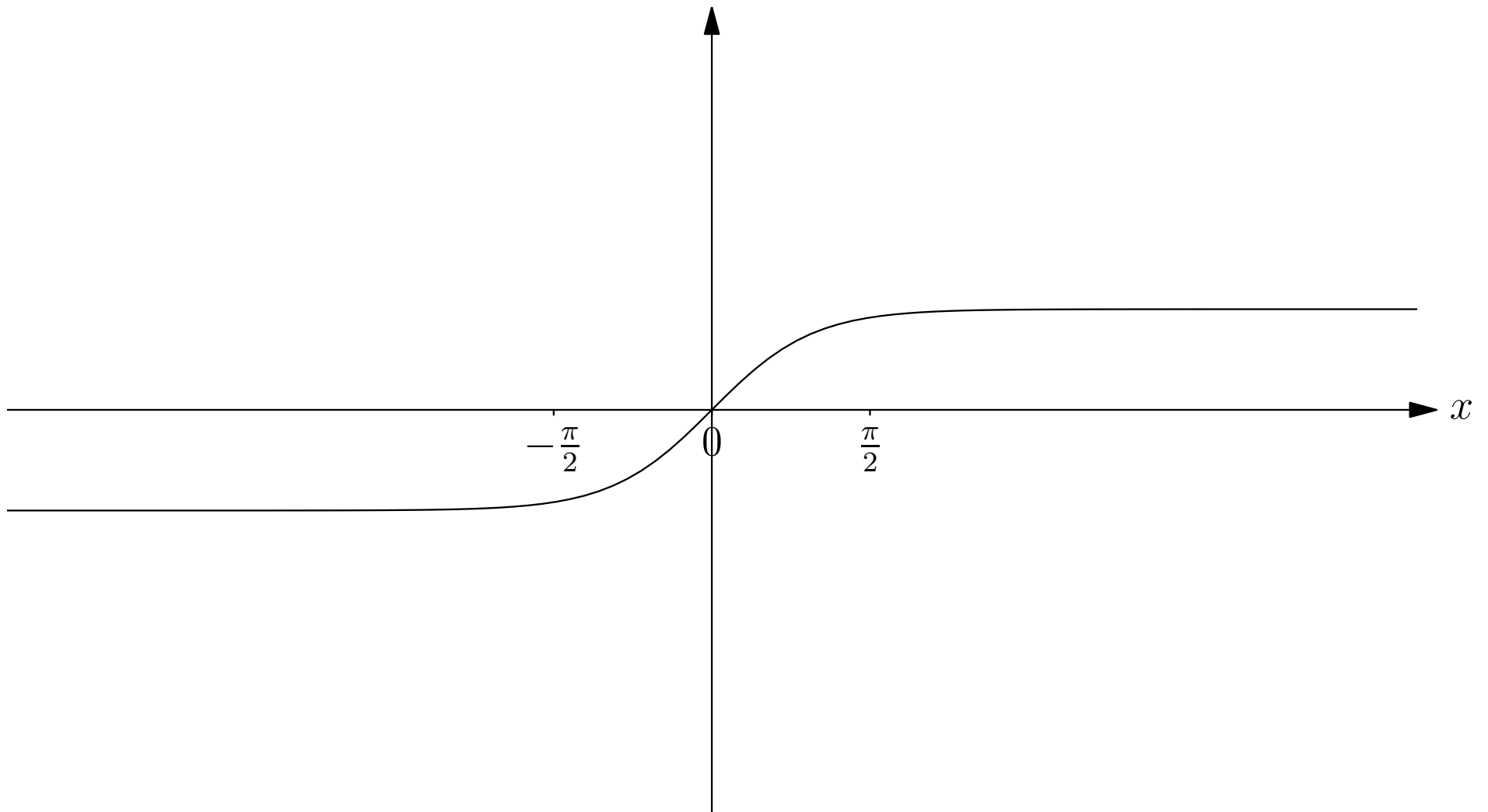
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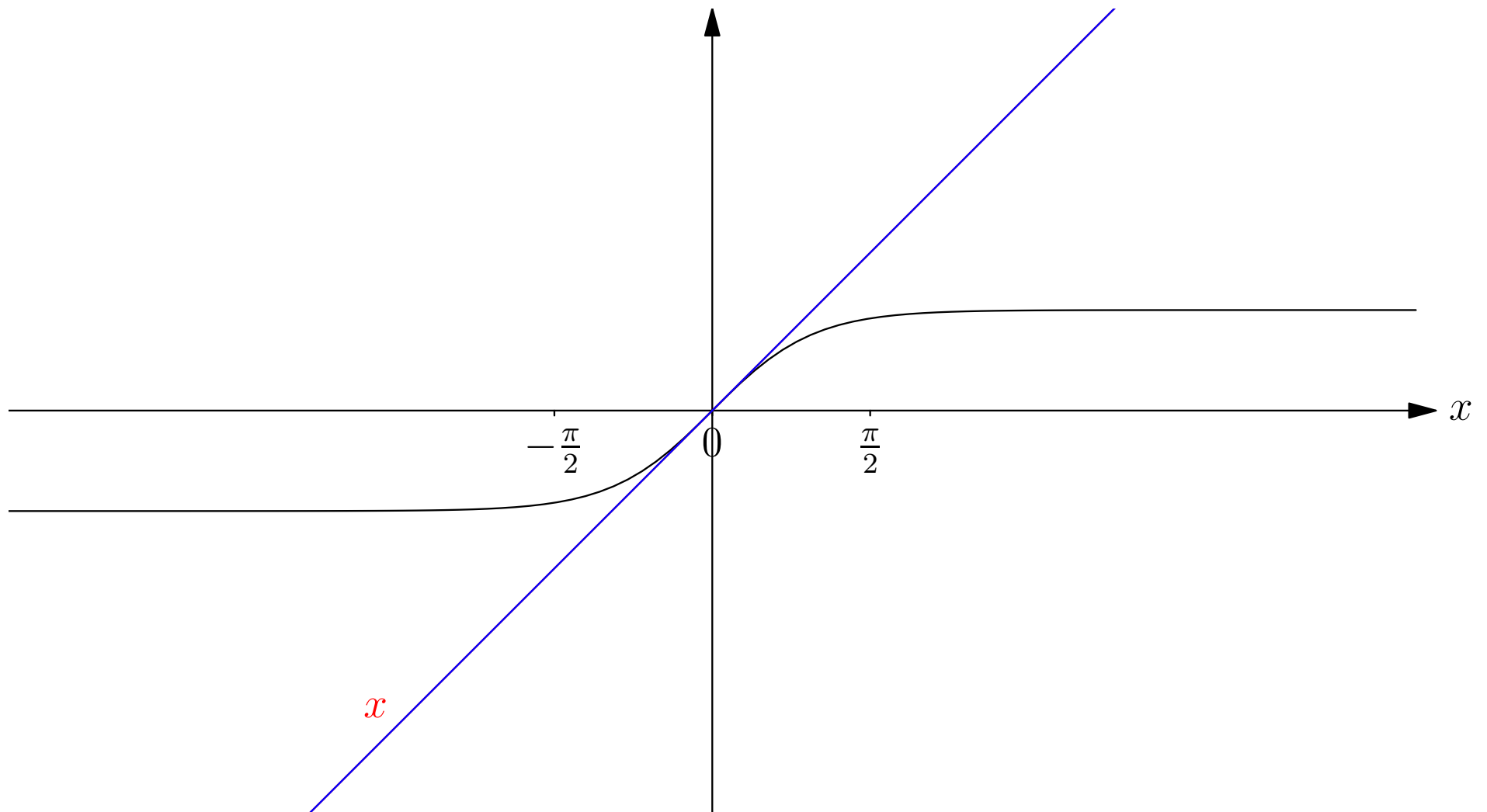
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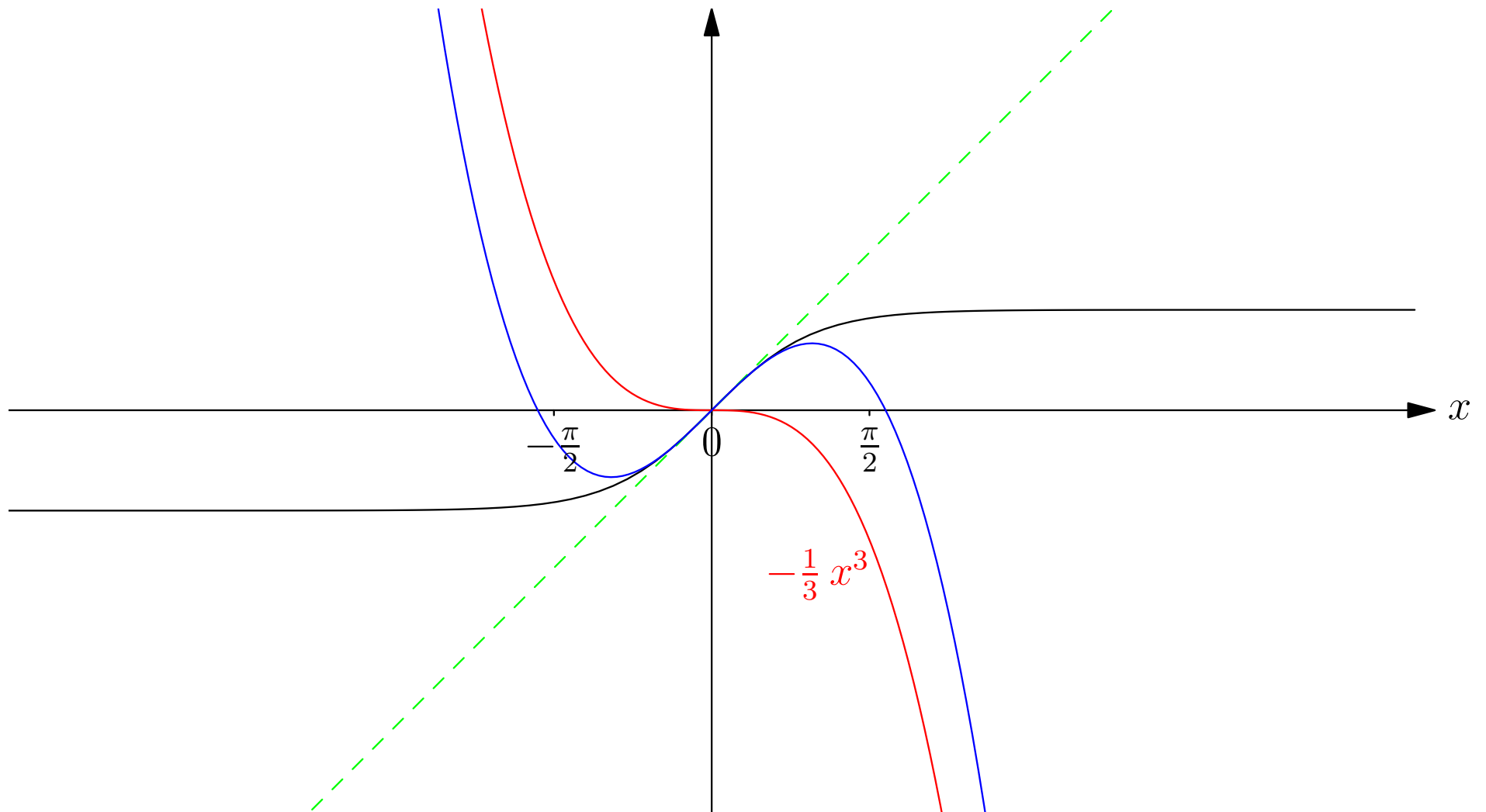
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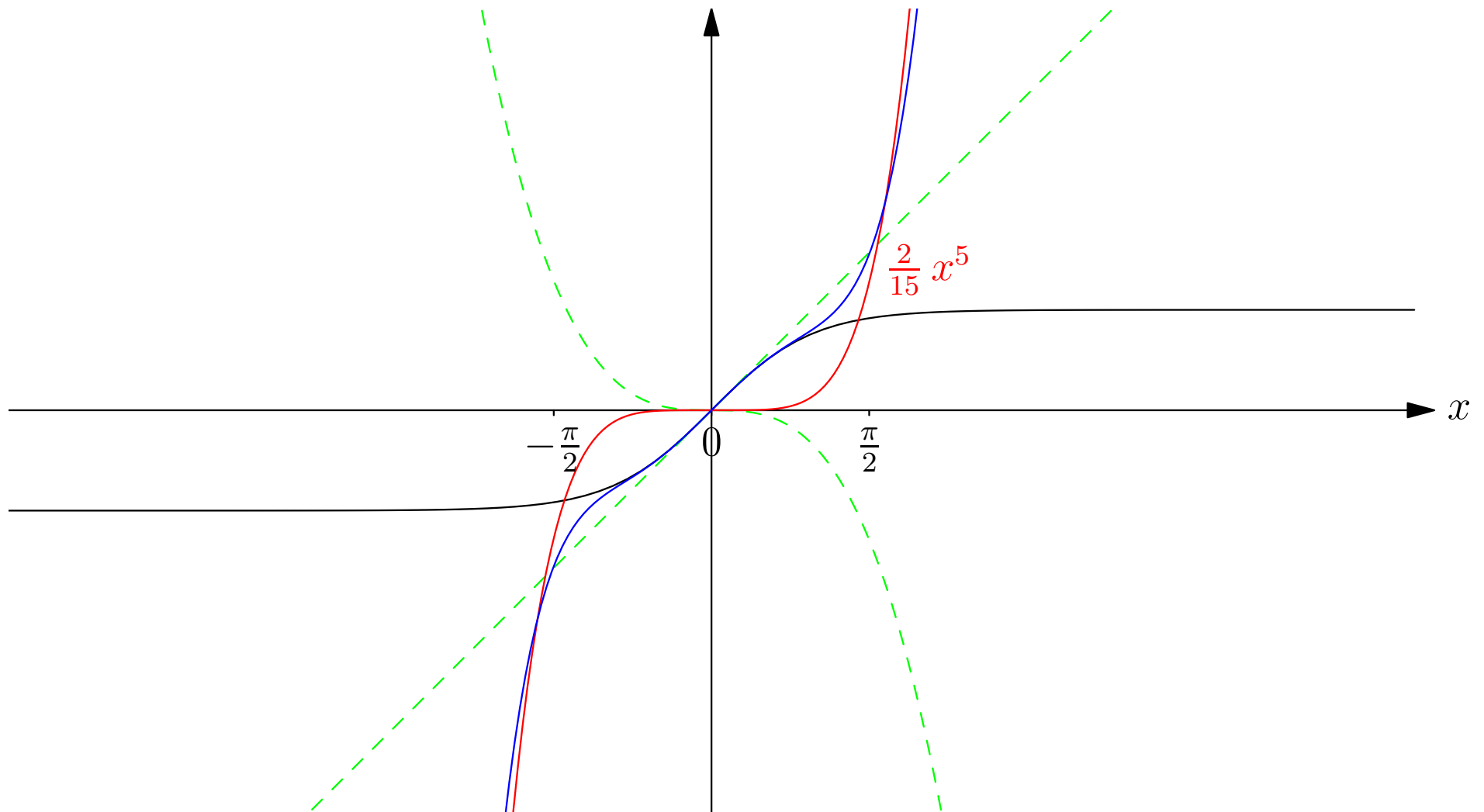
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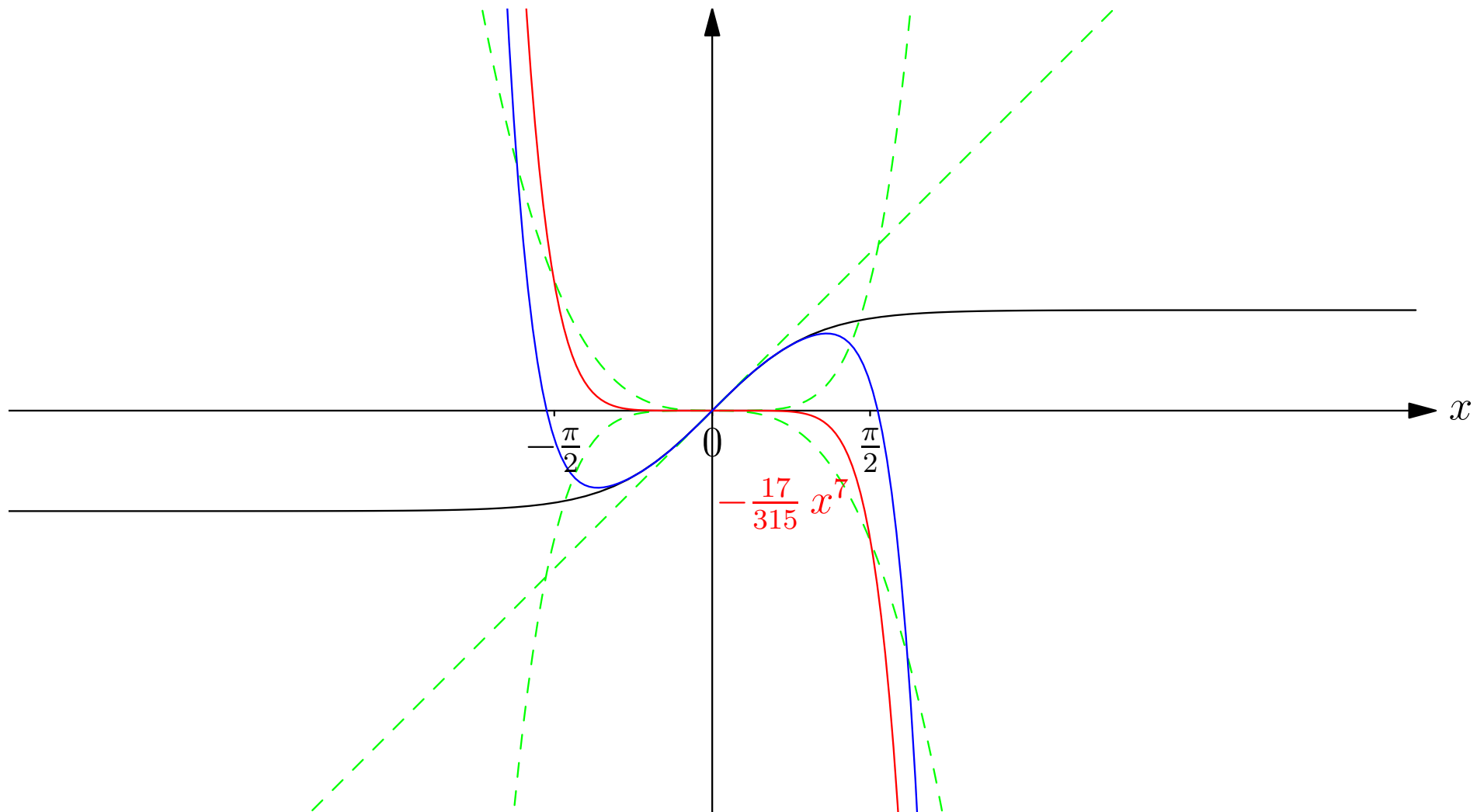
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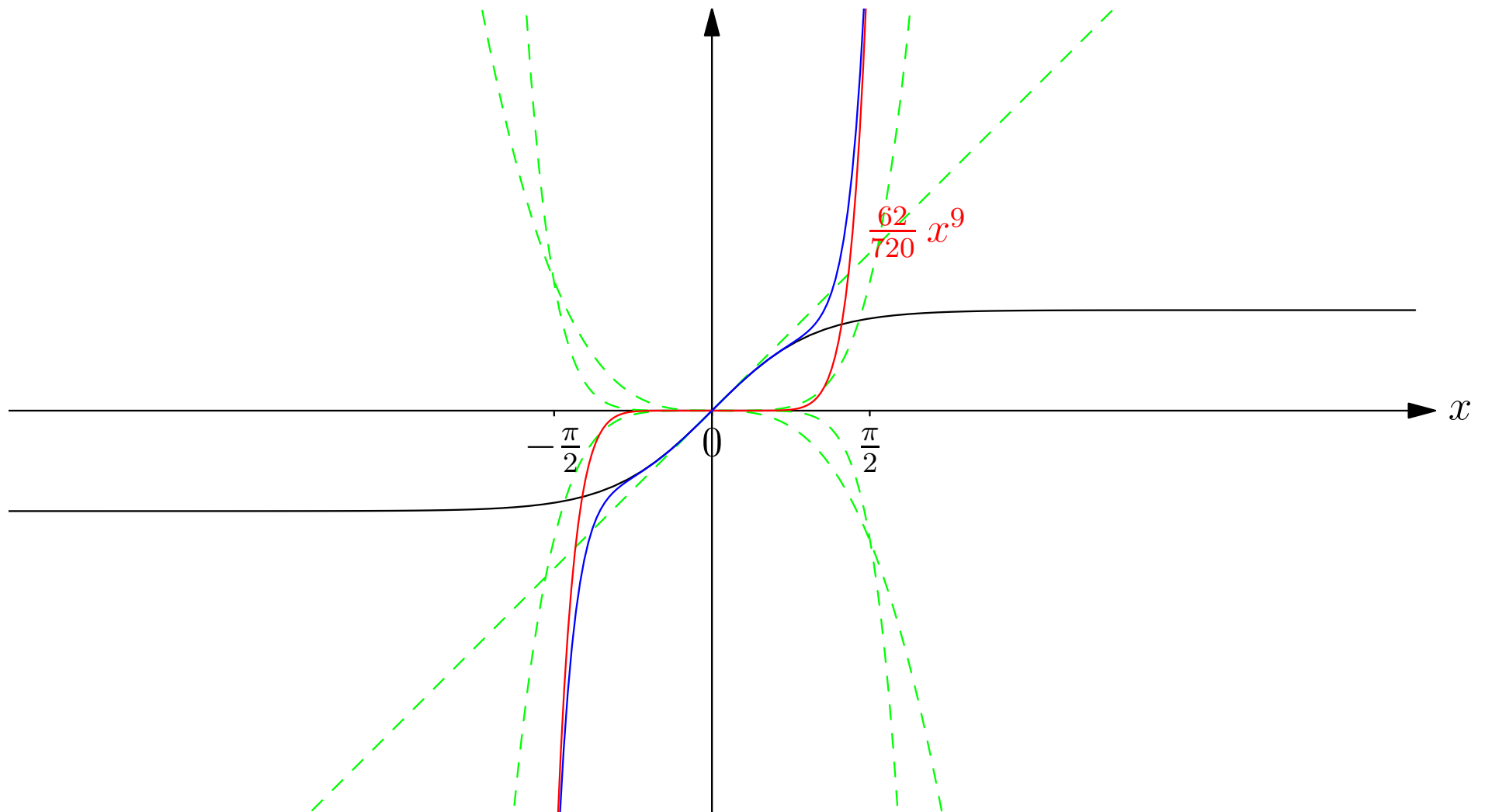
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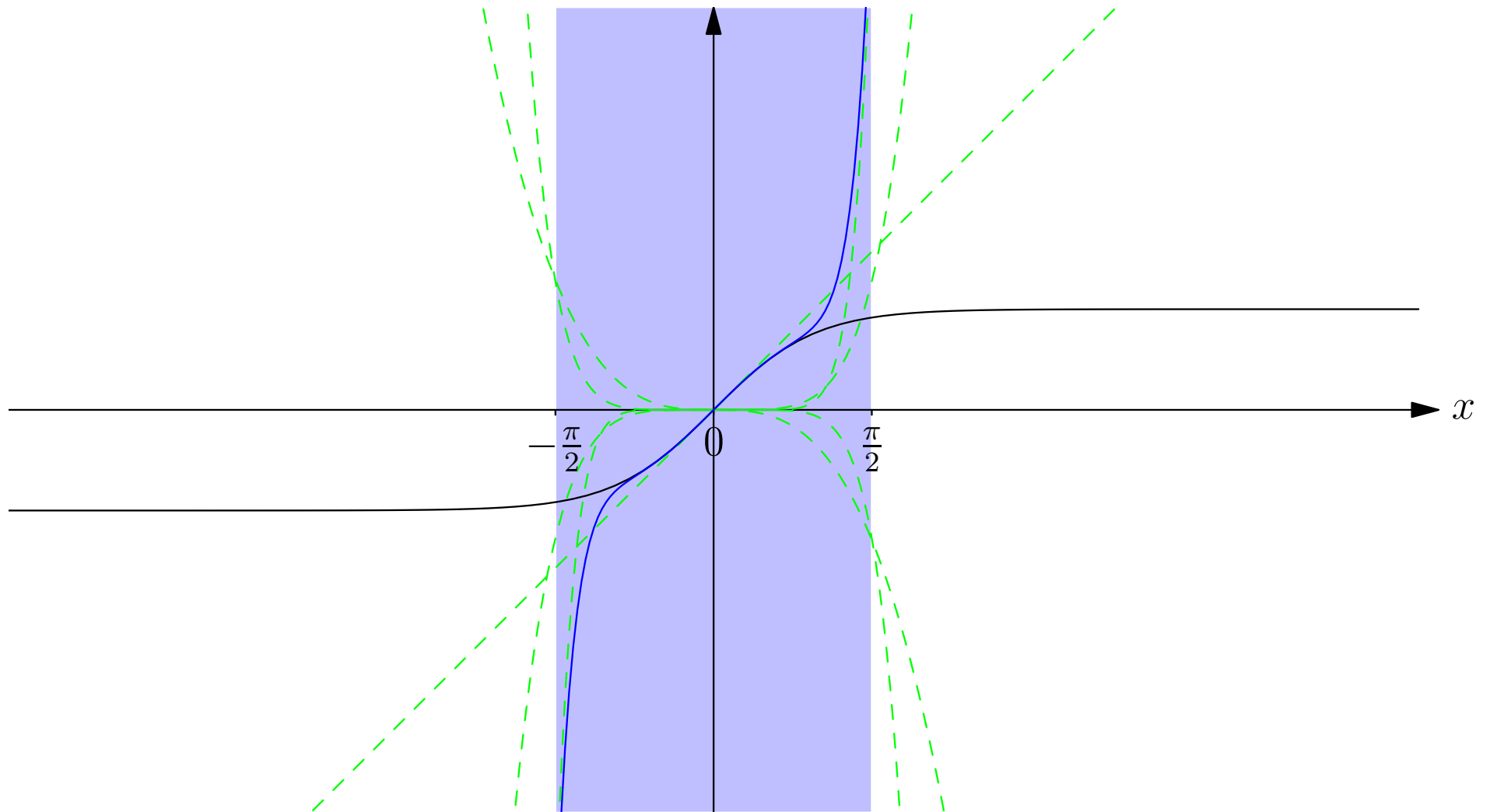
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Remainder

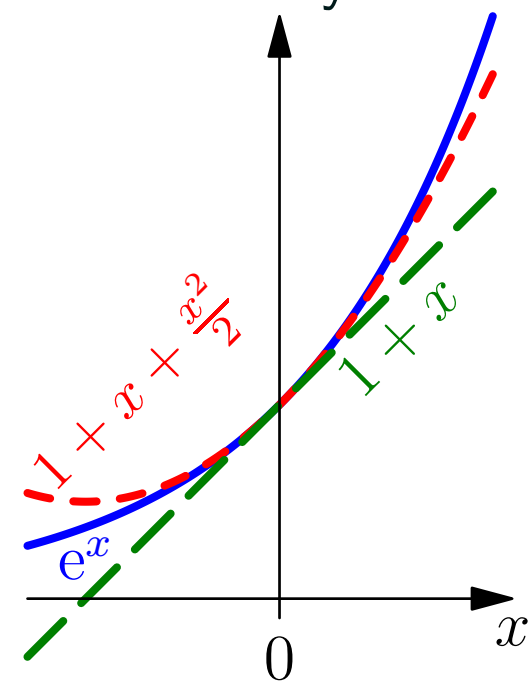
- Taylor expansions actually come with an error bound

$$f(x) = \sum_{i=0}^{n-1} \frac{(x - x^*)^i}{i!} f^{(i)}(x^*) + \frac{(x - x^*)^n}{n!} f^{(n)}(\theta_x)$$

θ_x is a number in the interval between x^* and x

- The last term, known as the **remainder**, can often be very useful

$$e^x = 1 + x + \frac{x^2}{2} e^{\theta_x}$$
$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} e^{\theta_x}$$



Remainder

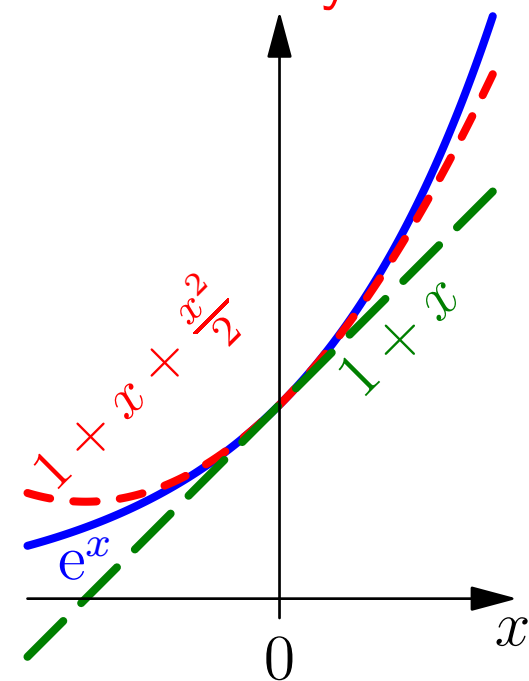
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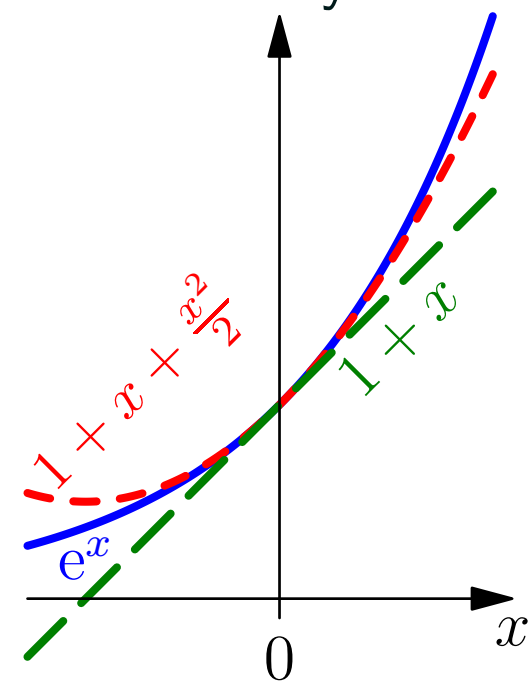
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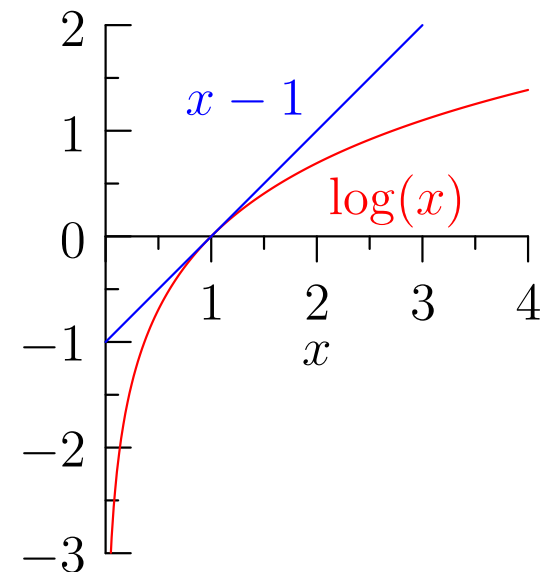
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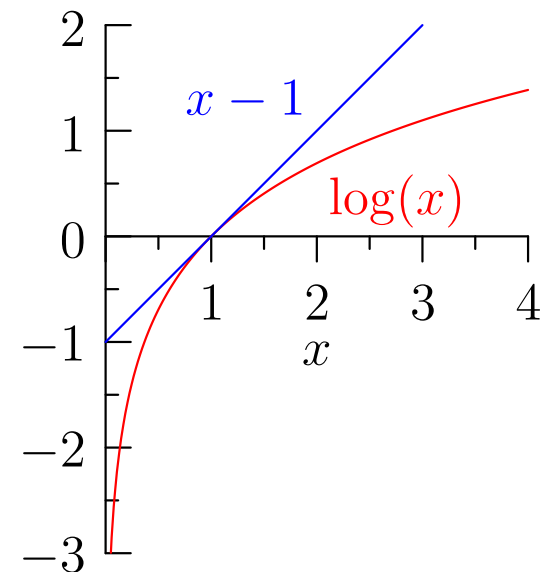
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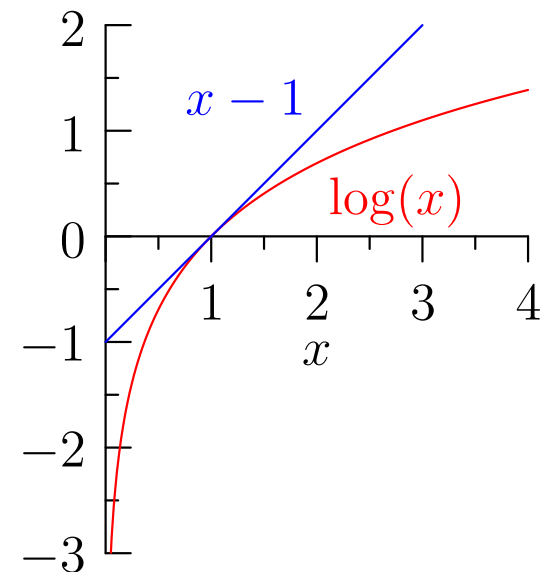
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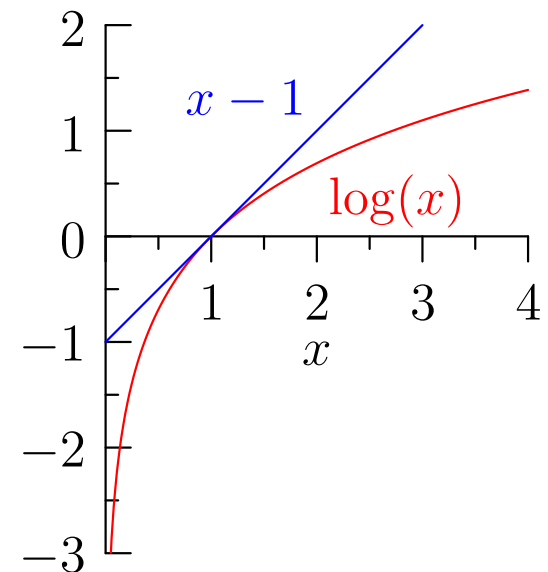
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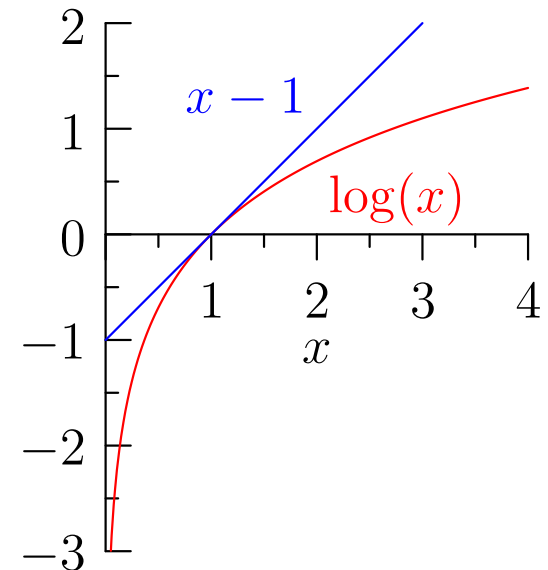
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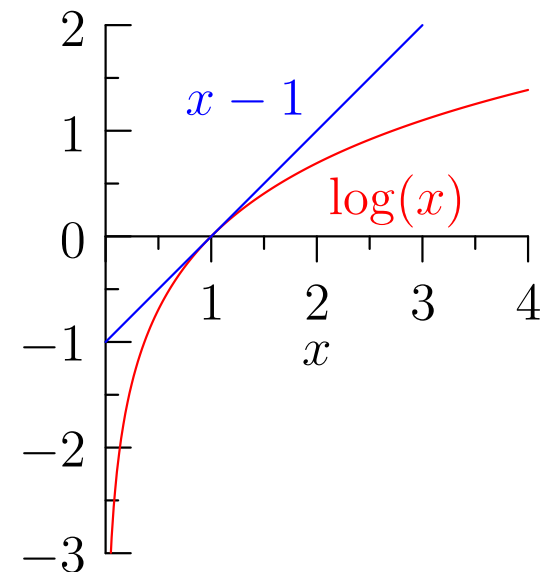
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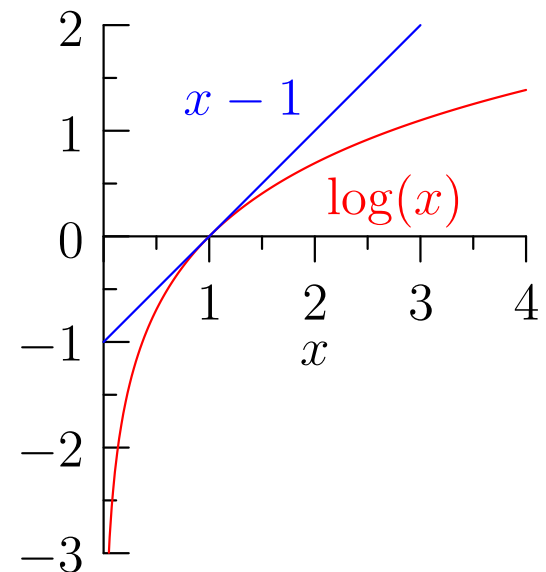
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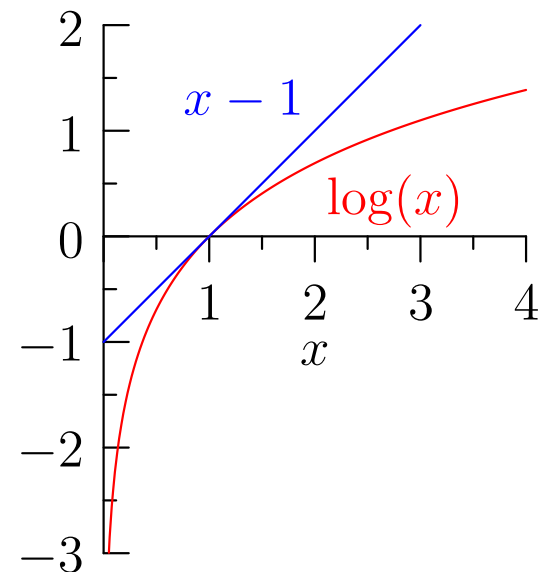
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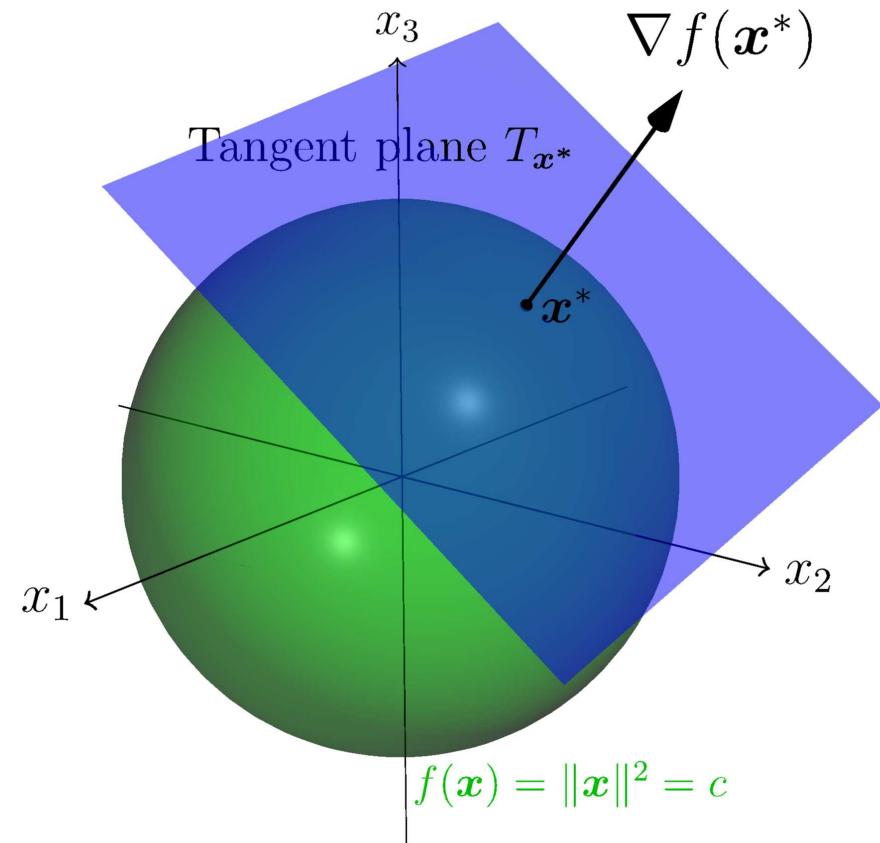
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Outline

1. 1-D Taylor Expansion
2. **High Dimensions**
3. Hessians



High Dimensional Spaces

- Just as we defined a Taylor expansion by approximating the function by a polynomial we can do the same for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(\mathbf{x}) \approx c^{(0)} + \sum_{i=1}^n c_i^{(1)} (x_i - x_i^*) + \sum_{i,j=1}^n (x_i - x_i^*) c_{ij}^{(2)} (x_j - x_j^*) + \dots$$

- We can determine the coefficients by differentiating both sides and setting $\mathbf{x} = \mathbf{x}^*$

$$f(\mathbf{x}^*) = c^{(0)}$$

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High Dimensional Taylor Expansion

- We find

$$f(\mathbf{x}) \approx f(\mathbf{x}^*) + (\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + \dots$$

- Where the matrix $\mathbf{H}$ is known as the Hessian and has components

$$H_{ij} = \frac{\partial^2 f(\mathbf{x}^*)}{\partial x_i \partial x_j}$$

- Because

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

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the Hessian is symmetric ($\mathbf{H} = \mathbf{H}^\top$)

Tangent Plane

- For smooth functions in n dimensions the equation $f(\mathbf{x}) = c$ defines an $(n - 1)$ -dimensional hyper-surface
- This can be thought of as the high dimensional analogue of a contour line
- The plane that goes through a point $\mathbf{x}^*$ that lines up with the contour surface is known as the tangent plane
- Away from $\mathbf{x}^*$ the value of the function $f(\mathbf{x})$ differs from the tangent plane by a quantity of order $\|\mathbf{x} - \mathbf{x}^*\|^2$

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Defining the Tangent Plane

- Thus if $\mathbf{x}$ is in the tangent plane $T_{\mathbf{x}^*}$ then

$$f(\mathbf{x}) = f(\mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^2)$$

- Or $(\mathbf{x} - \mathbf{x}^*)^\top \nabla f(\mathbf{x}^*) = 0$
- That is the tangent plane at a point $\mathbf{x}^*$ is the set of points orthogonal to the gradient

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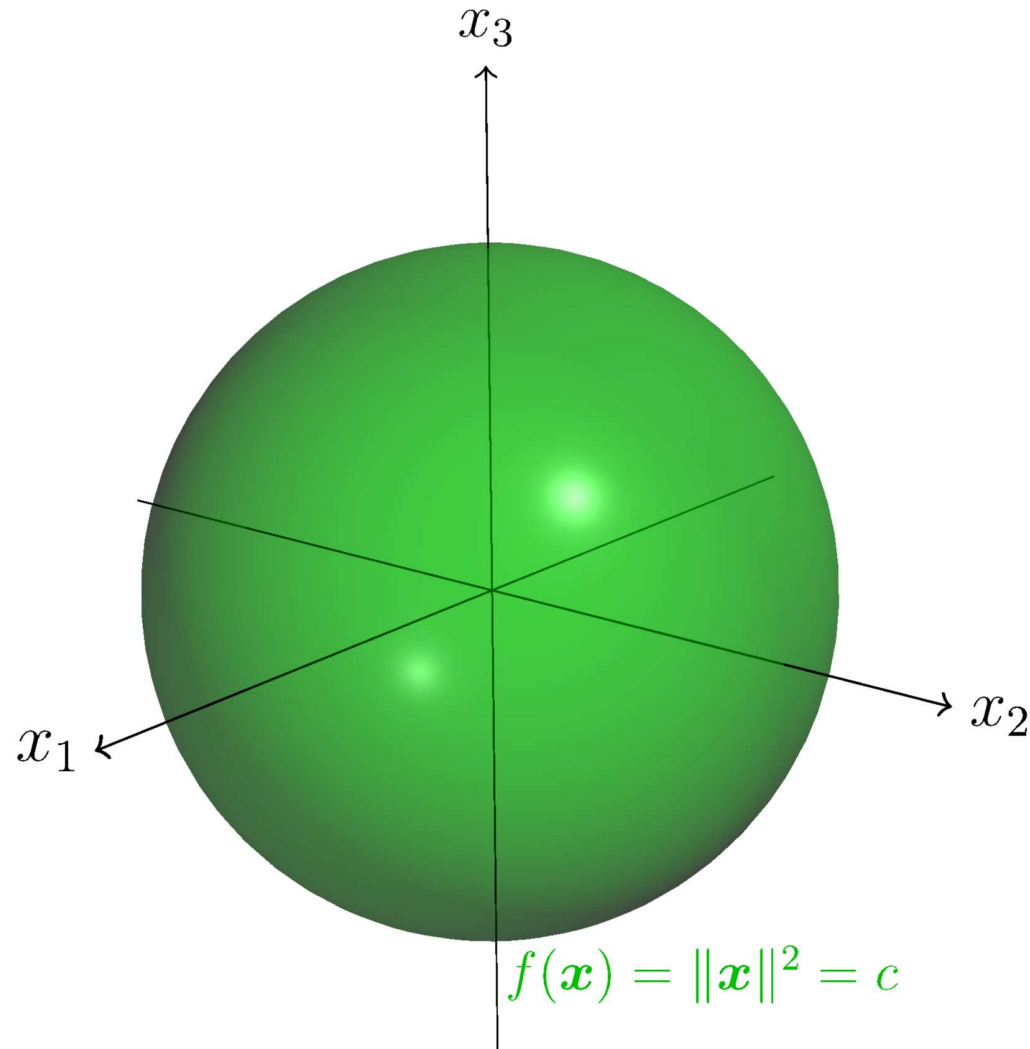
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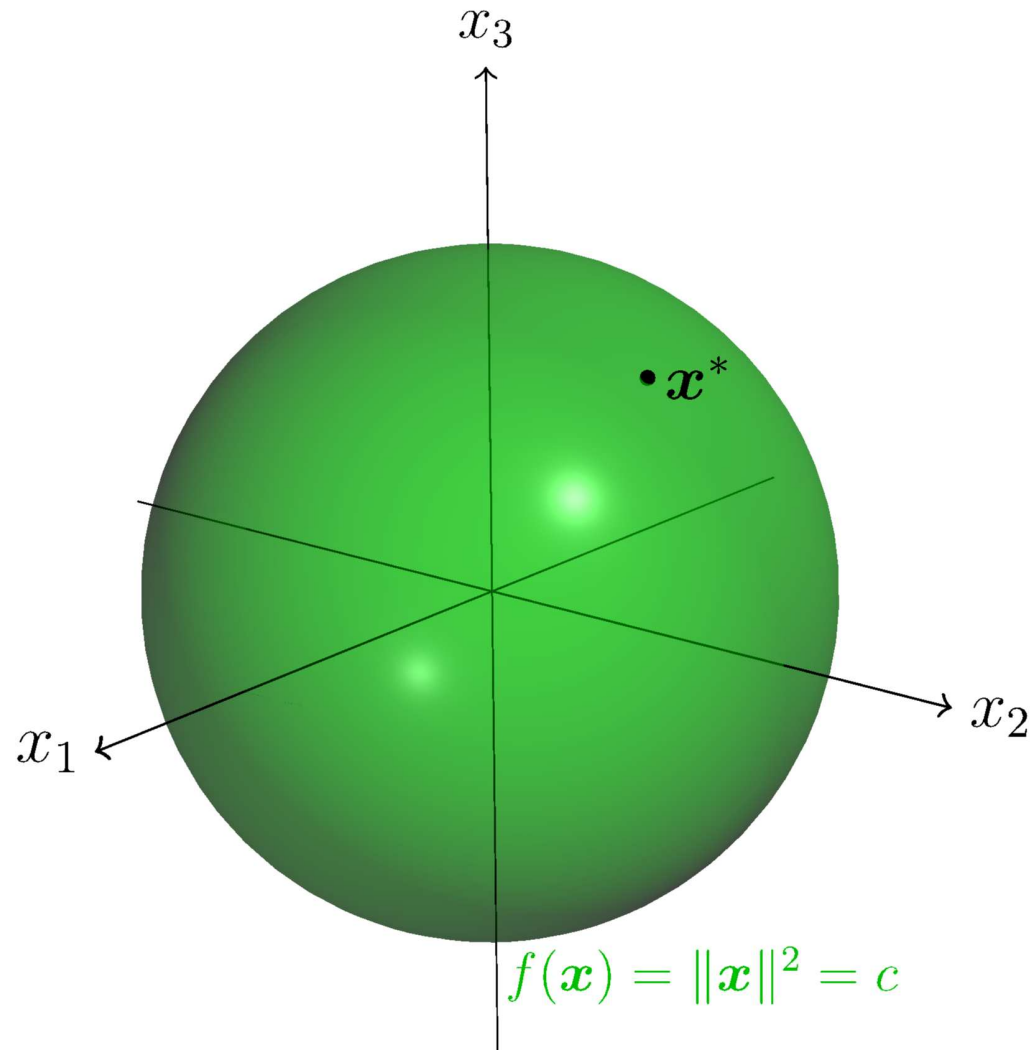
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- Notice that the gradient then points in the direction orthogonal to the tangent plane of the contour surface

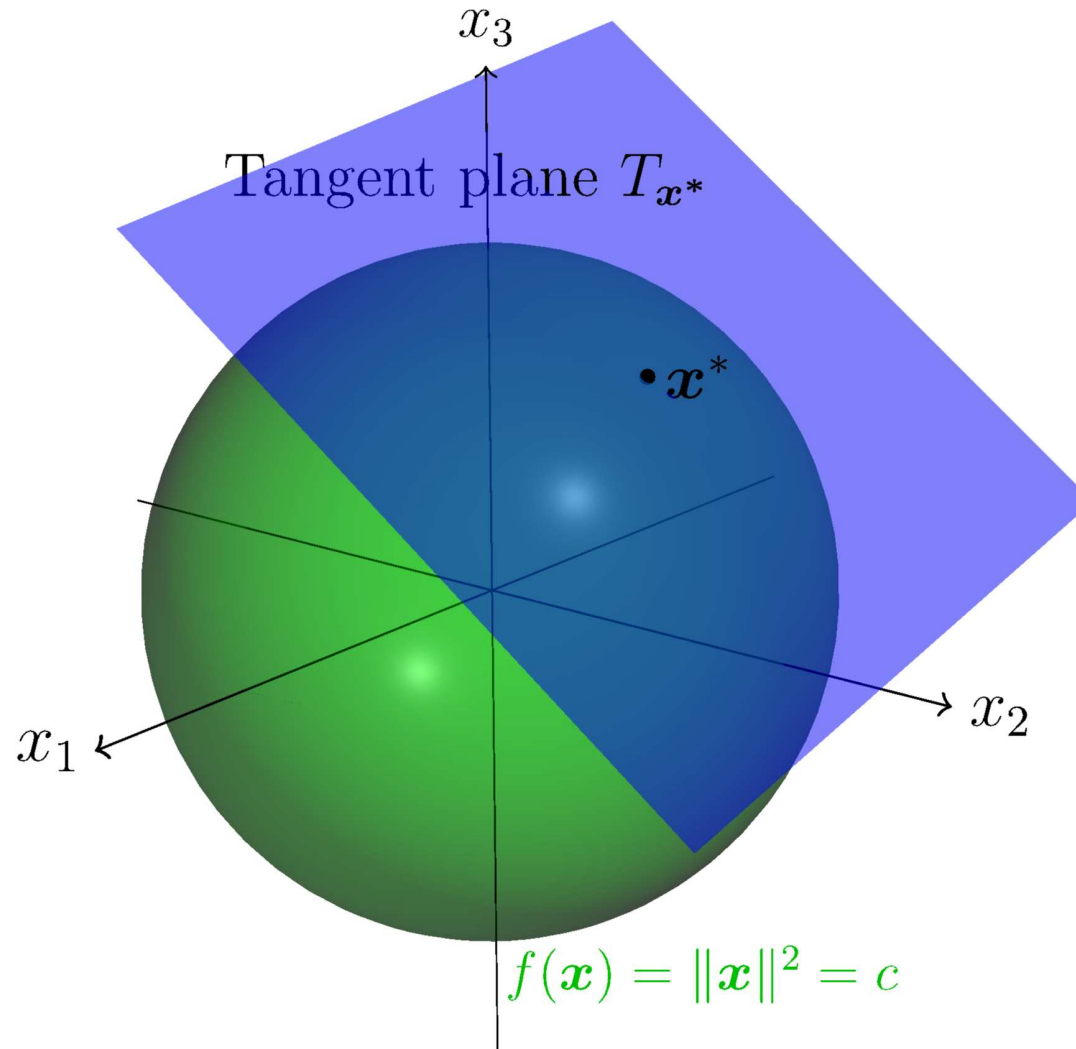
Picture of Tangent Plane for $f(\mathbf{x}) = \|\mathbf{x}\|^2 = c$



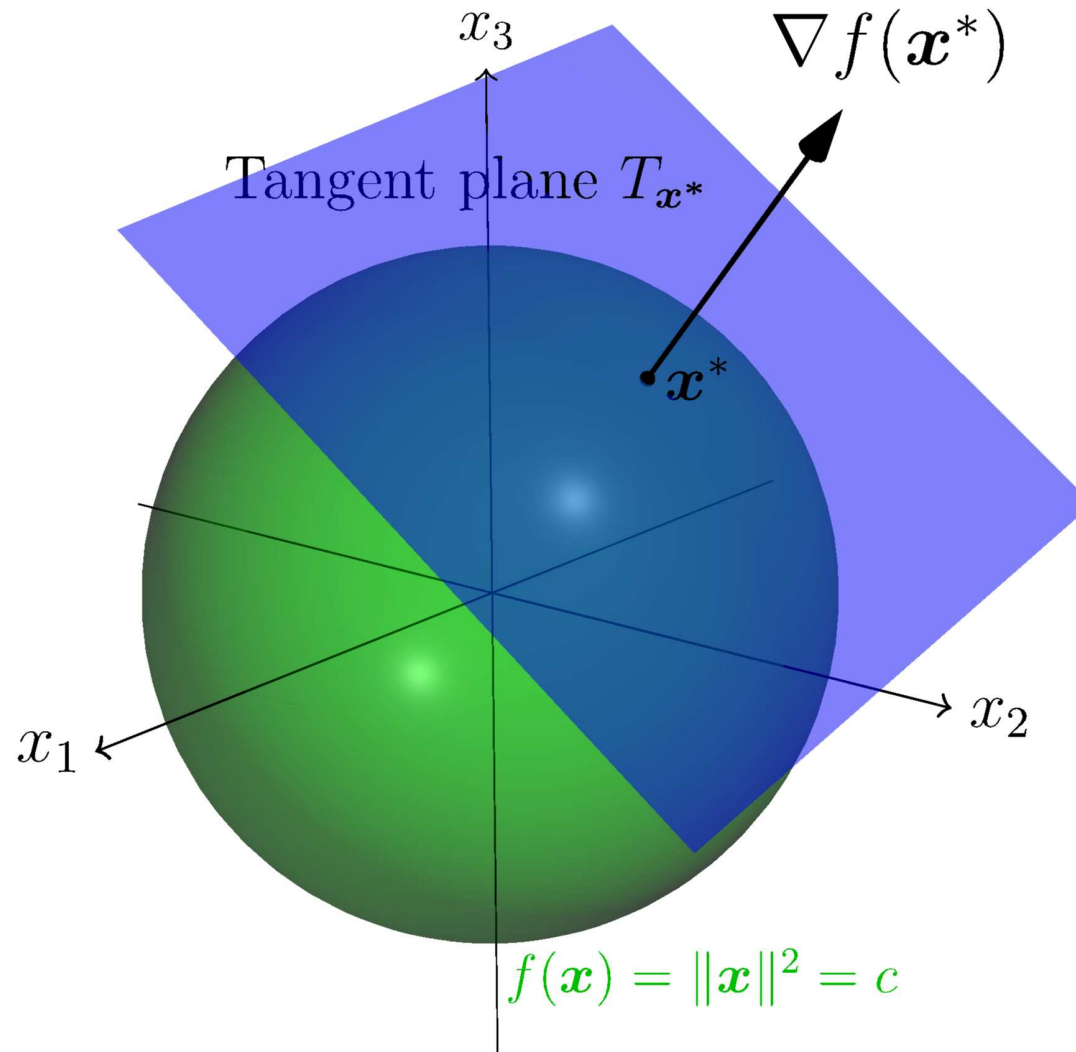
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Direction of Steepest Ascent

- We note that as

$$f(\mathbf{x}^* + \epsilon \mathbf{u}) = f(\mathbf{x}^*) + \epsilon \mathbf{u}^\top \nabla f(\mathbf{x}^*) + O(\epsilon^2)$$

- For a step $\mathbf{u}$ of fixed length, the function locally grows fastest when $\mathbf{u}$ is in the direction $\nabla f(\mathbf{x}^*)$
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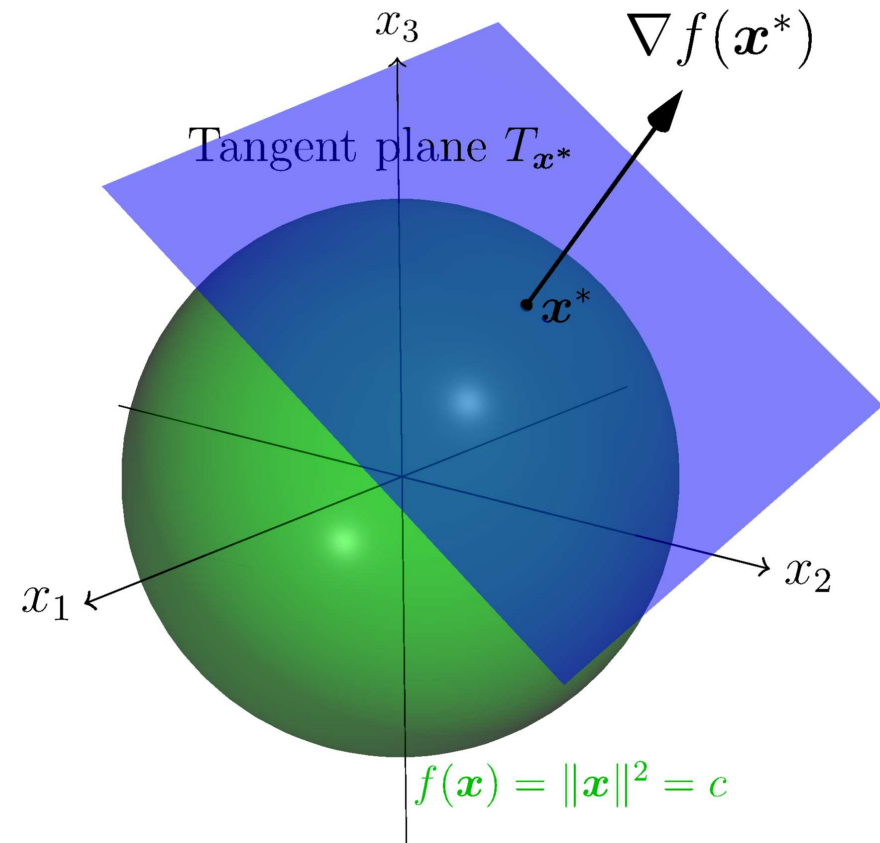
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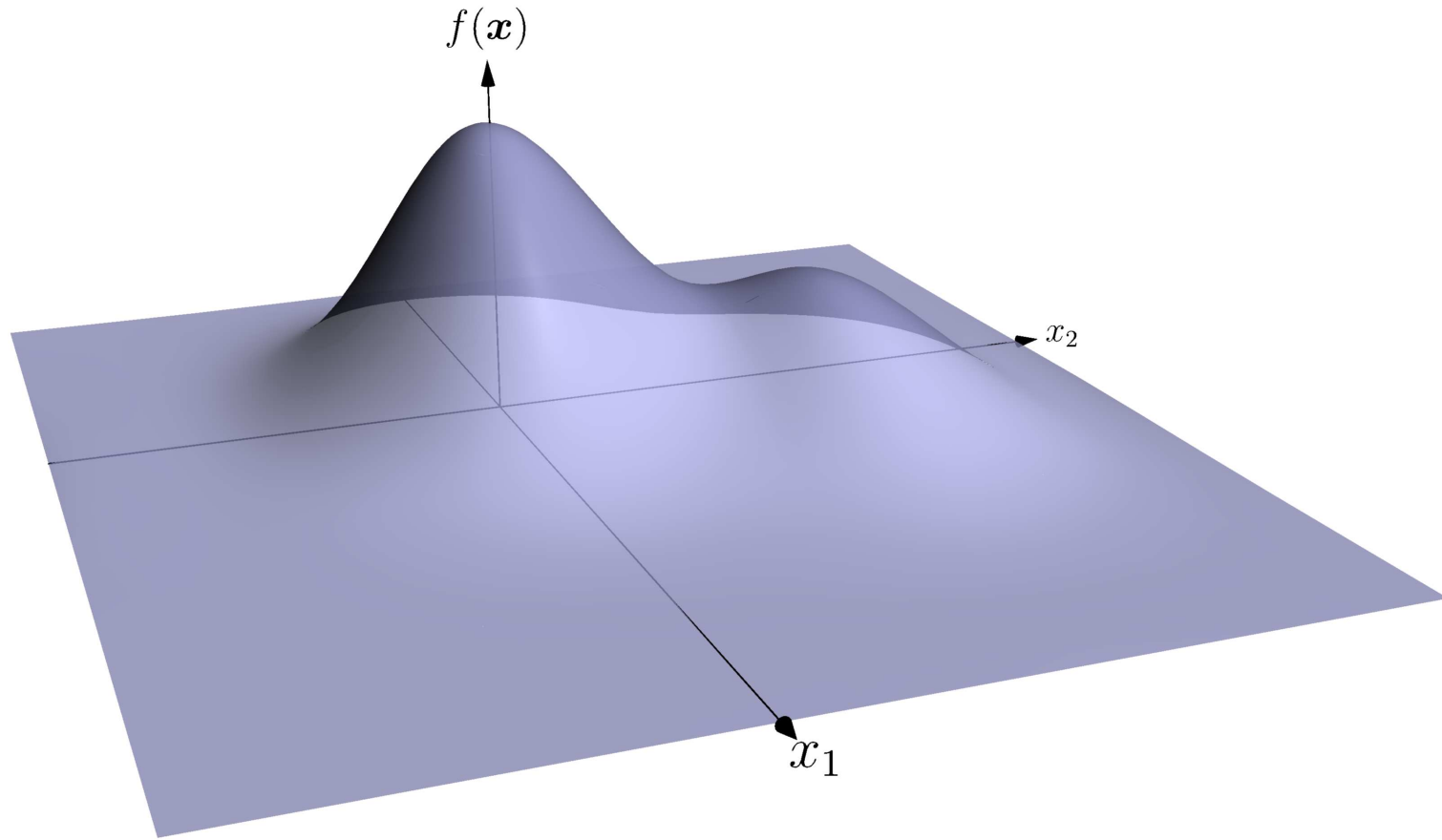
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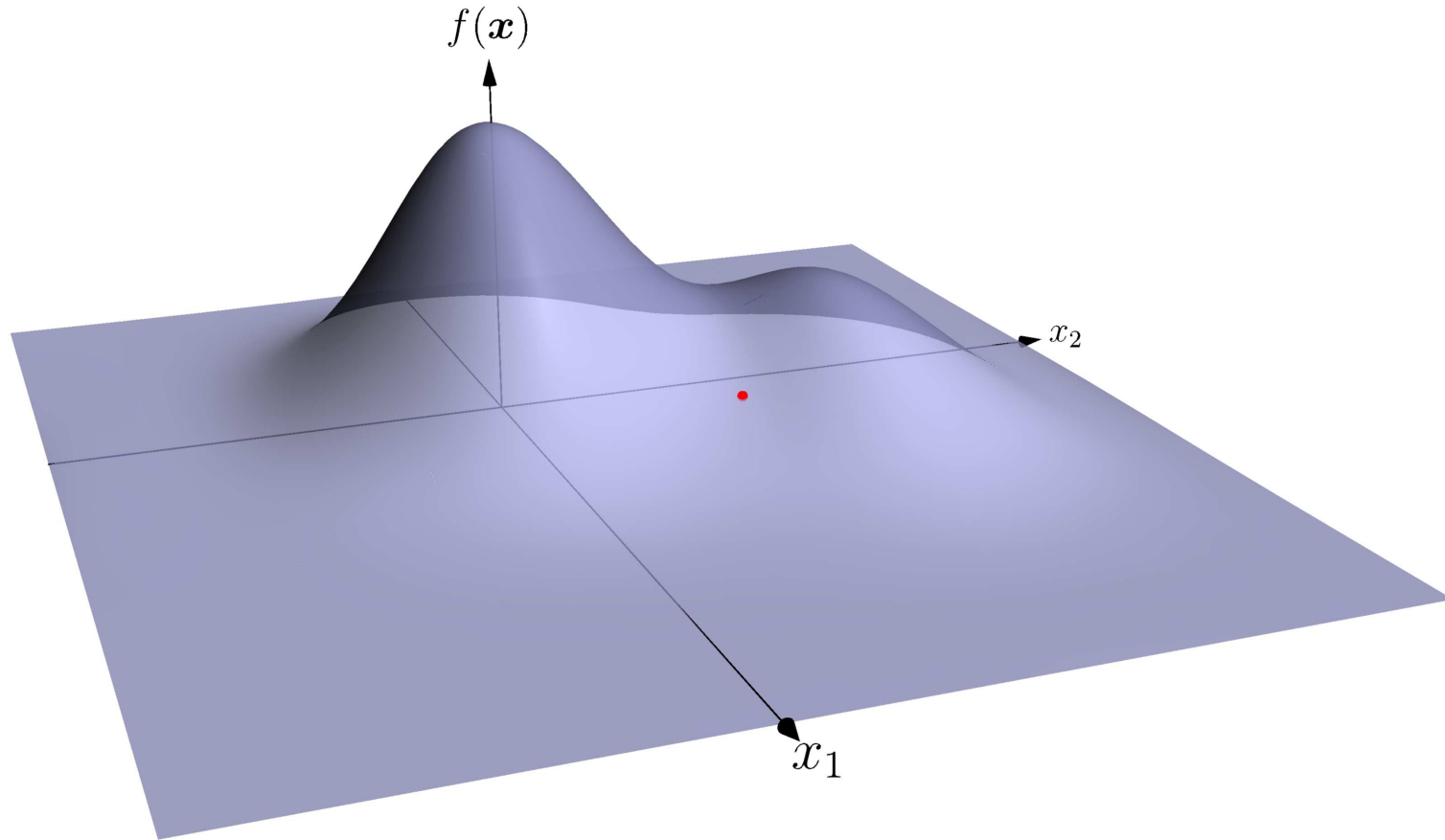
1. 1-D Taylor Expansion
2. High Dimensions
3. **Hessians**



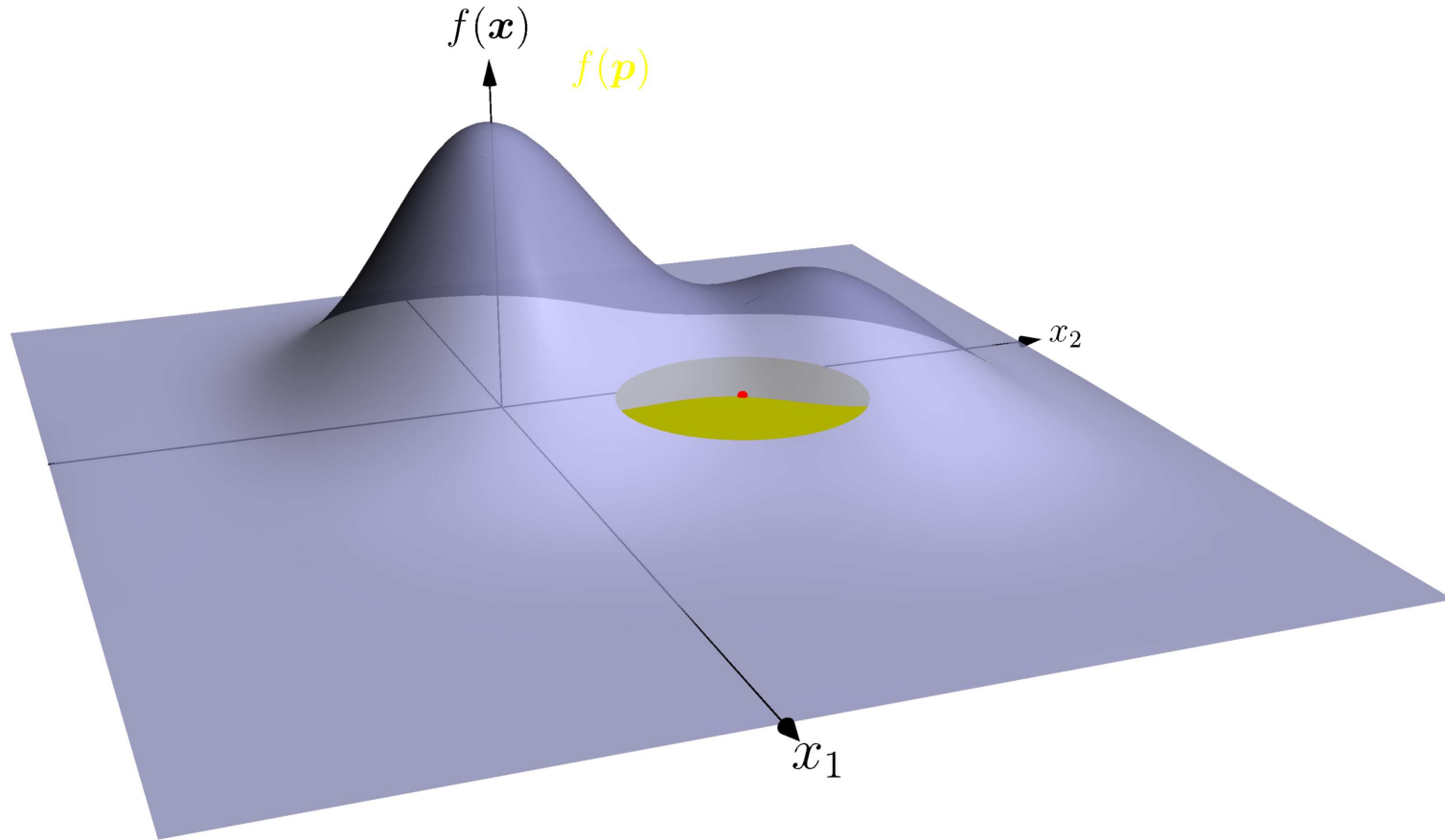
2-D Taylor Expansion



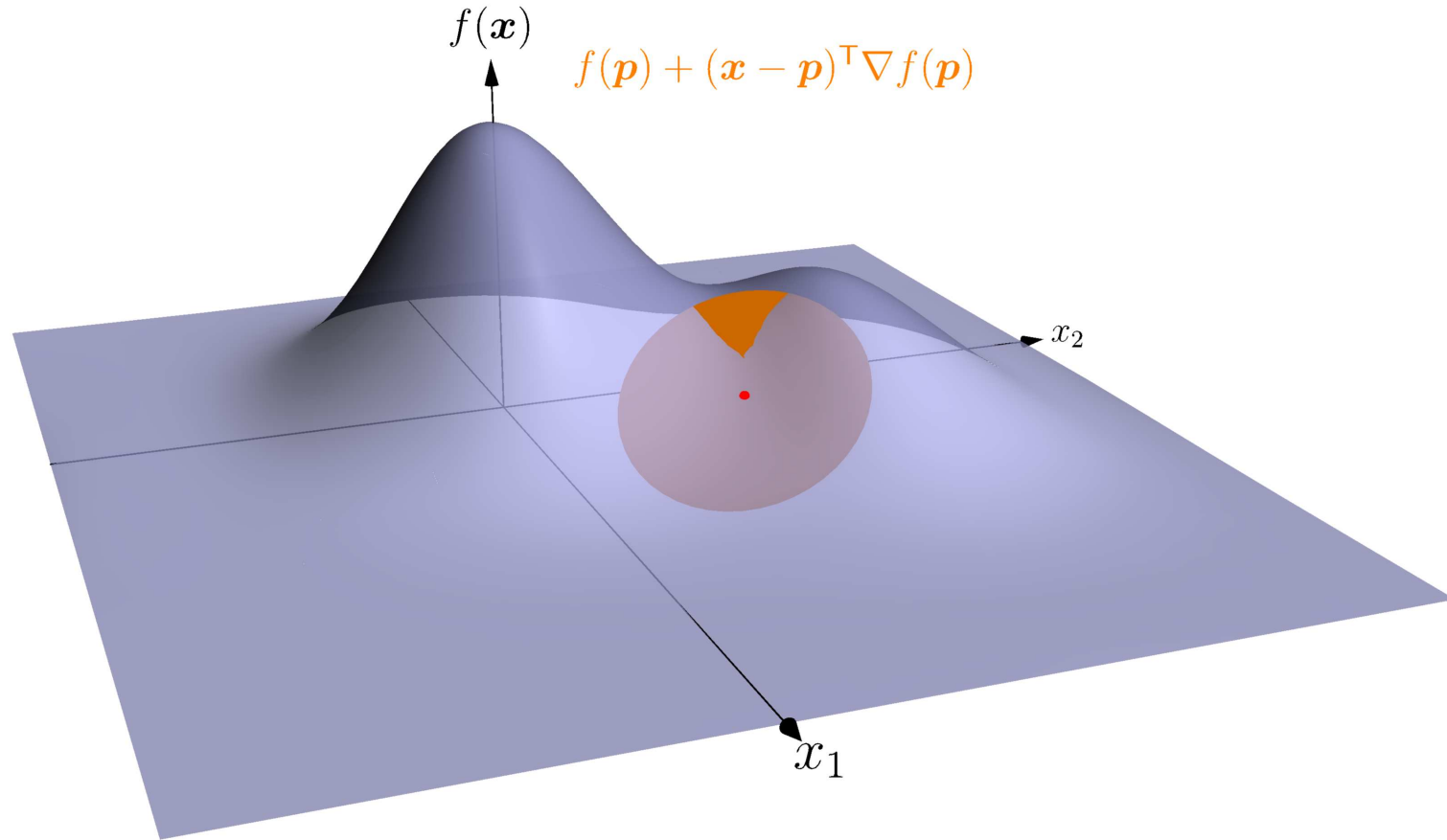
2-D Taylor Expansion



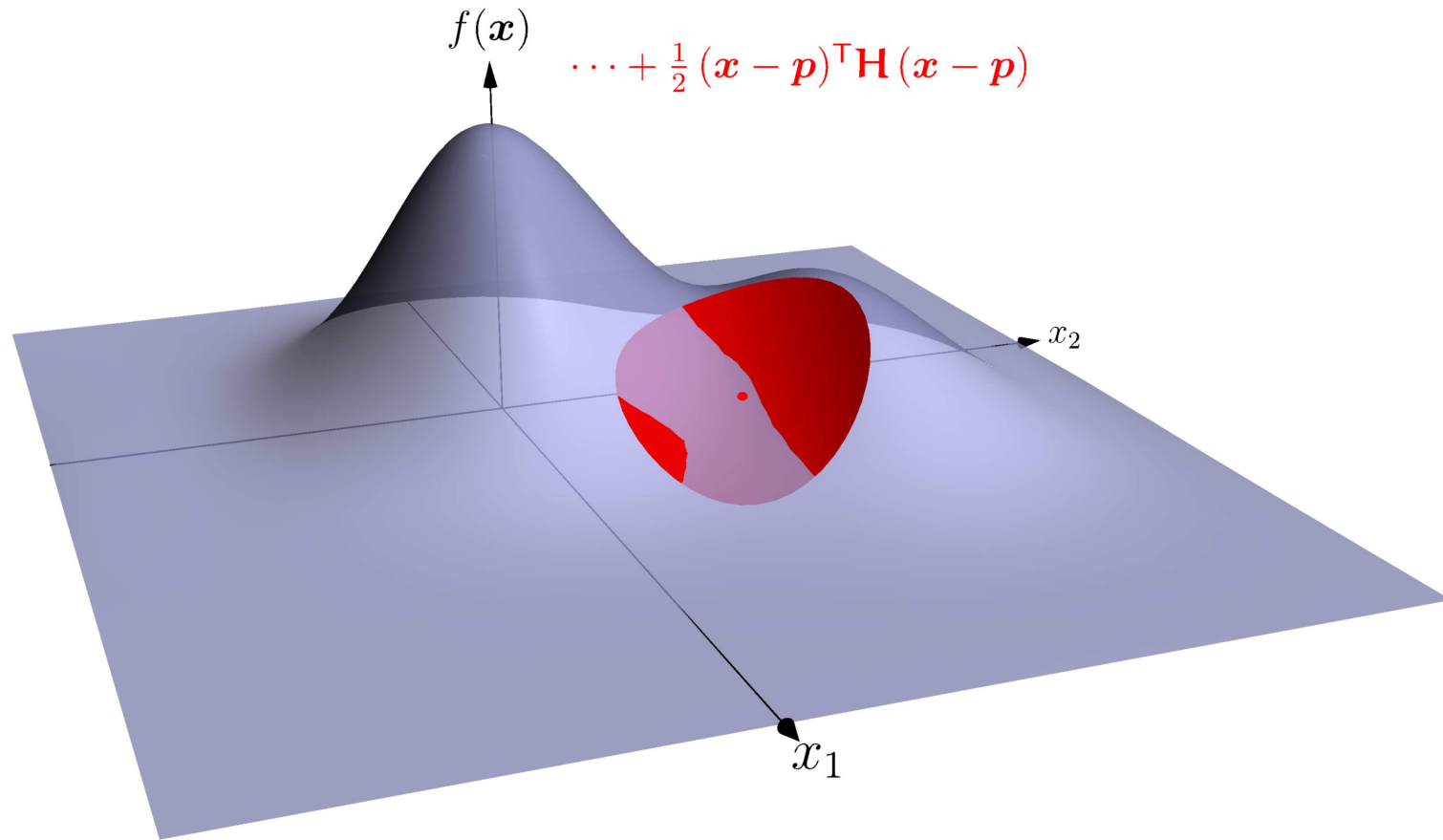
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The Hessian

- The second order term in the Taylor expansion tells us about the **curvature** of the function
- This is very relevant when we are at a **stationary point**, $\mathbf{x}^*$, where $\nabla f(\mathbf{x}^*) = \mathbf{0}$ so that

$$f(\mathbf{x}) = f(\mathbf{x}^*) + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{H}(\mathbf{x} - \mathbf{x}^*) + O(\|\mathbf{x} - \mathbf{x}^*\|^3)$$

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Eigenvalues

- We will revisit eigen-systems, but recall that for a square matrix an eigenvector, $\mathbf{v}_i$, of a matrix $\mathbf{H}$ is defined such that

$$\mathbf{H}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

- For an $n \times n$ symmetric matrix (such as the Hessian), the eigenvalues are all real, and we can choose n orthonormal eigenvectors such that $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$
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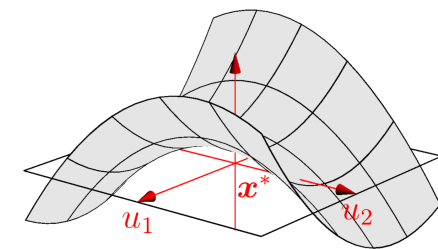
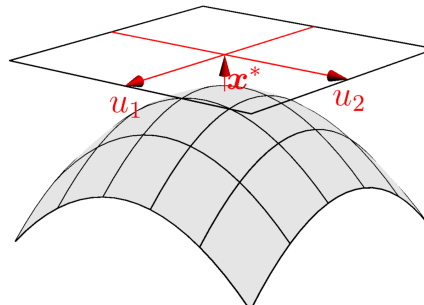
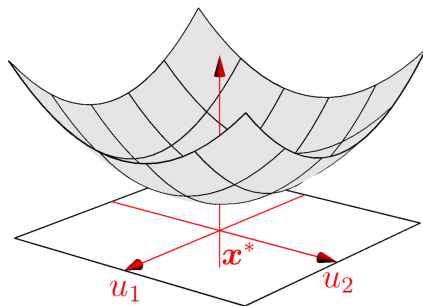
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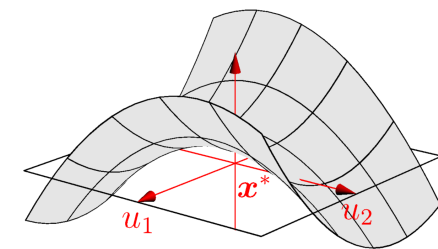
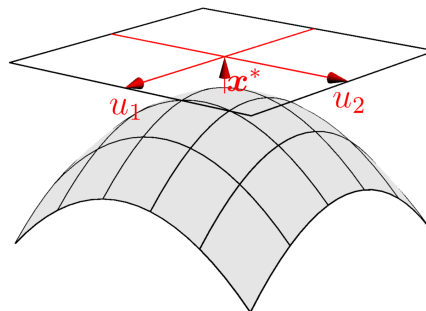
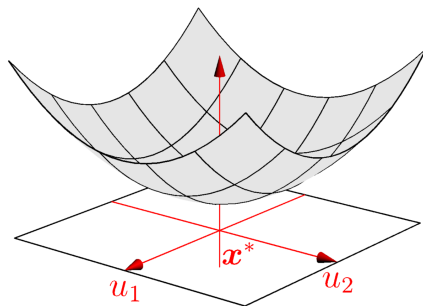
Minima, Maxima and Saddle-Points

- If all the eigenvalues are positive, $\lambda_i > 0$, then for any u we have $u^T H u > 0$ and x^* would be a local minimum
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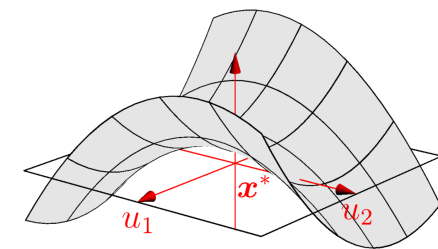
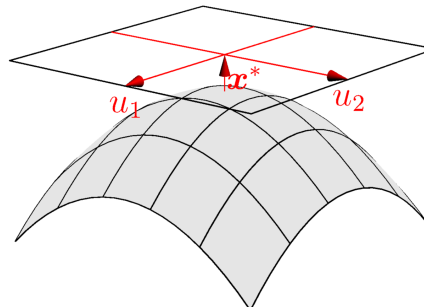
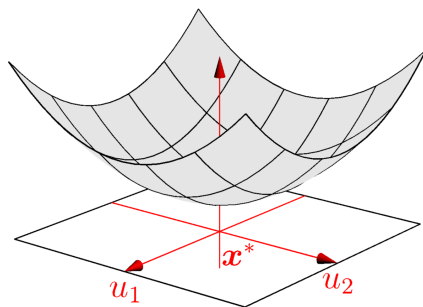
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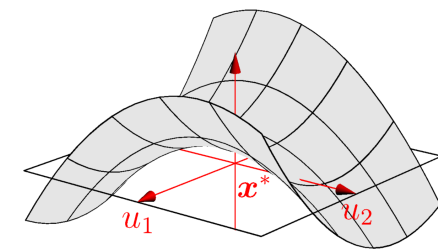
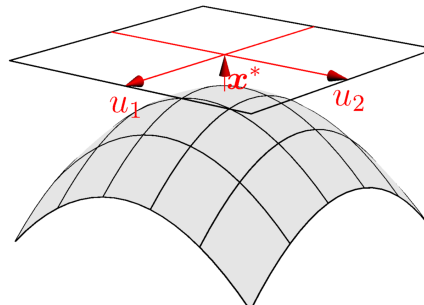
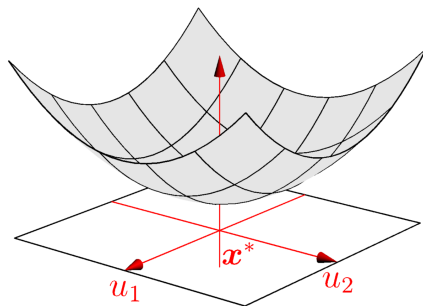
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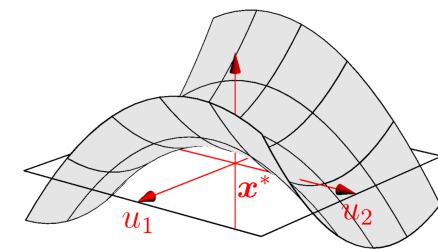
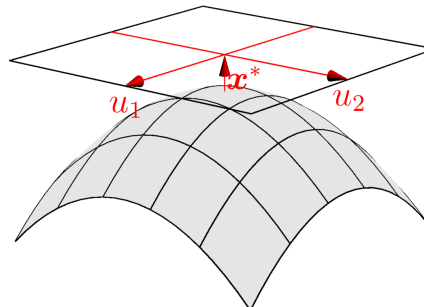
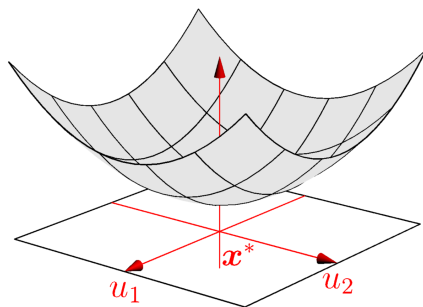
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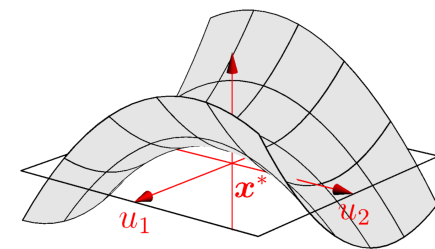
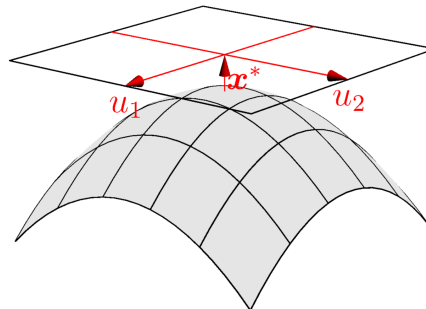
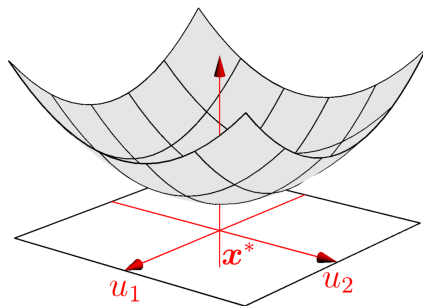
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Finding Optima

- We are often given a loss function, $L(\mathbf{w})$, which we wish to minimise
- A sufficient condition for $\mathbf{w}^*$ to be a local minimum is that $\nabla L(\mathbf{w}^*) = \mathbf{0}$
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